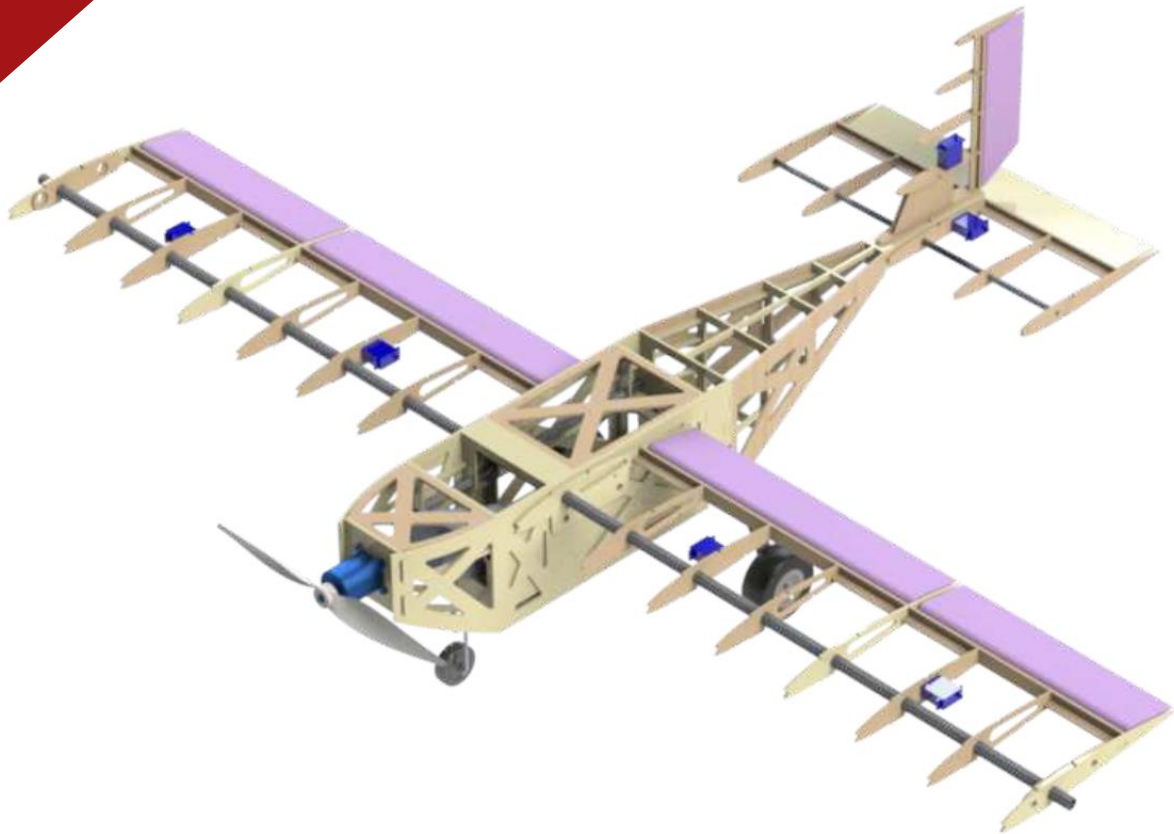




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Nomenclature

Abbreviations

AIAA	American Institute of Aeronautics and Astronautics	RC	Radio-controlled
AoA	Angle of attack	TOFL	Takeoff field length
AR	Aspect ratio	T-O	Takeoff
AUW	All up weight	VLM	Vortex lattice method
CA	Cyanoacrylate	VT	Vertical Tail
CAD	Computer-aided design	WU	Washington University
CBDM	Component drag build-up method	WUDBF	Washington University in St. Louis Design/Build/Fly
CF	Carbon fiber	Symbols	
CFD	Computational Fluid Dynamics	b	Span [m]
CG	Center of gravity	c	Chord [m]
DBF	Design/Build/Fly	C_o	Drag coefficient
ESC	Electronic Speed Controller		
FEA	Finite Element Analysis	C_L	Lift coefficient
GM	Ground Mission	Re	Reynolds number
HT	Horizontal Tail	S	Wing Area [m ²]
HQRS	Cooper-Harper handling qualities rating scale	T	Thrust [kgf]
KM Sim	Kinematic mission simulator	T/W	Thrust-to-weight ratio
LiPo	Lithium Polymer	V_{HT}	Horizontal tail volume
M1	Mission 1	v_{max}	Max cruise velocity [m/s]
M2	Mission 2	V_{VT}	Vertical tail volume
M3	Mission 3	W	Weight [kgf]
NACA	National Advisory Committee for Aeronautics	W/S	Wing loading [kgf/m ²]
NiMH	Nickel-metal hydride		

1. Executive Summary

The Washington University in St. Louis Design/Build/Fly (WUDBF) 2022-23 Design Report describes the design, testing, and manufacturing of *Bear Force One*, the team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2023 Design/Build/Fly competition. For the competition, the aircraft must be able to complete three flight missions and one ground mission. The first flight mission (M1) is a staging flight that verifies the function of the aircraft. The second mission (M2) is a surveillance flight; the aircraft must carry an Electronics Package payload while traversing the course as fast as possible. The third mission (M3) is a jamming flight which involves flying three timed laps while carrying a Jamming Antenna externally mounted to a wing. The ground mission (GM) serves as a structural margin demonstration of the aircraft.

Bear Force One has been designed to meet all mission requirements while maintaining a high level of reliability and minimizing risk. The aircraft consists of a conventional, single motor, bow tricycle landing gear, high-wing design. The bow tricycle configuration was chosen for its structural stability and because, due to the lenient 60 ft TOFL requirement, it was determined that no takeoff angle of attack would be necessary. The high-wing configuration maximizes stability because it keeps the aircraft's CG below the wings. A sensitivity analysis performed during the conceptual design phase indicated that a M2 payload of 1.22 kg and an M3 antenna length of 0.86 m was optimal given the size and scoring constraints.

During the preliminary design phase, the aircraft was further refined via detailed aerodynamic and propulsion design. Lift, drag, and stability analyses were performed to size the aircraft and set the outer geometry. The wingspan was set at 1310 mm with a maximum wing loading of roughly 11 kg/m². The M2 payload placement was chosen such that the change in overall CG between missions is minimized. A propulsion system was then selected that gives the aircraft a 1.38 thrust-to-weight ratio and a predicted maximum cruise velocity of 37.78 m/s.

During the detail design phase, the internal structure of the aircraft was defined. The M3 antenna and ground fixture adapters were designed in conjunction to maintain a consistent attachment point with the wing. The rear of the fuselage was designed primarily for optimizing access to the M2 payload.

Based on subsequent performance simulations, the aircraft was predicted to be able to take off in less than 33 ft. For M2, the aircraft was predicted to be able to complete 17 laps in the 10-minute window yielding a score of 1.46. The aircraft was predicted to complete M3 in 1.82 minutes yielding a score of 2.42. These predicted flight times aligned well with endurance estimates made in the conceptual design phase. Upon summing these predicted scores with the predicted M1 and GM scores, *Bear Force One* is estimated to achieve a total score of 5.87 out of a possible 7. Thorough ground and flight testing confirmed that the aircraft could meet all predicted performance goals.

2. Management Summary

2.1 Organization

To facilitate efficient communication and work, WUDBF is organized into an Administrative Branch and a Technical Branch, both of which are overseen by a Faculty Advisor. Information and decisions flow downward from the Presidents to the Team Leads and then ultimately to the General Members. The Administrative Branch is responsible for resource allocation, public relations, and recruiting. The Administrative Branch consists of the Administrative President, the Finance Lead, and the Communications Lead. The Technical Branch is responsible for the design and manufacture of the aircraft. The leadership structure of WUDBF is described in Figure 2.1. The Technical Branch consists of the Technical President and six Teams as described in Table 2.1. The number of General Members assigned to each Team is based primarily upon student interest, but it also depends on Team needs.

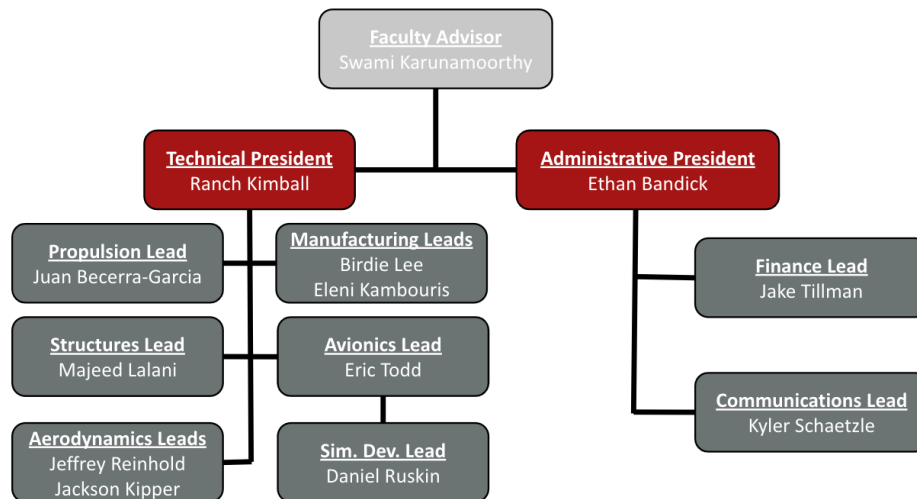


Figure 2.1: Team Hierarchy

Table 2.1: Team Roles and Skills

	Aerodynamics	Structures	Manufacturing	Avionics	Propulsion	Simulation Development
Roles	- Design of the exterior geometry of the aircraft - Aircraft sizing	Design and modeling of the aircraft structure	- Assembly of the aircraft - Materials testing	- Install of electrical components - Programming of flight controllers	- Design of the propulsion package - Motor testing	- Development of simulation tools - Sensitivity analysis
Skills	- Aircraft stability and control optimization - CFD analysis	- CAD - Structural FEA - Mass properties analysis	- Laser cutting 3D printing - Machining	- Electrical design - Soldering - Real Flight 8	- Propulsion theory and analysis - eCalc - RC Benchmark	- MATLAB - Simulink - Python - Real Flight 8

2.2 Schedule

WUDBF uses a Gantt chart to ensure the team stays on track to meet major deadlines. The Gantt chart for the 2022-2023 competition year is shown in Figure 2.2 below.

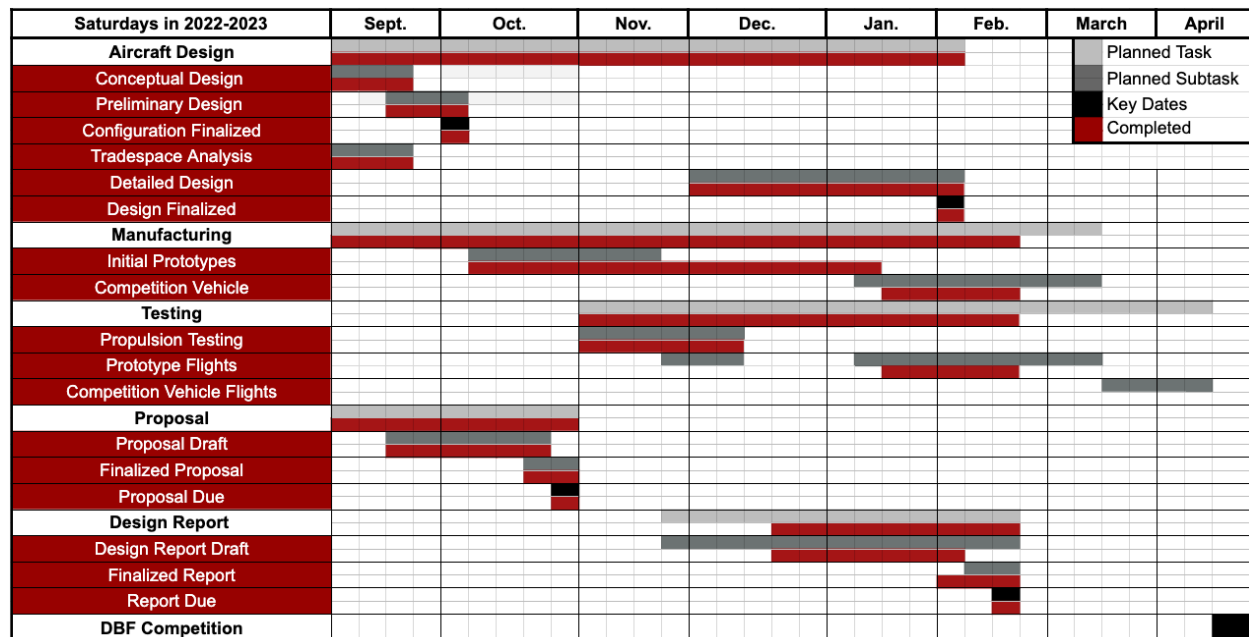


Figure 2.2: WUDBF Gantt Chart for 2022-2023 Competition

3. Conceptual Design

For the 2022-23 DBF competition, the AIAA has set the objective to design and build an electronic warfare aircraft. The aircraft must be capable of transporting an electronics package and externally carrying a jamming antenna while maintaining a compact design.

Upon receipt of the competition rules, WUDBF began the conceptual design phase starting with identification of key mission requirements and translation of those mission requirements into design requirements, followed by a scoring sensitivity analysis and configuration selection via a series of decision matrices.

3.1 Mission Requirements

The 2023 competition consists of four missions; one ground mission and three flight missions. The same course is used for all flight missions and is shown in the figure below.

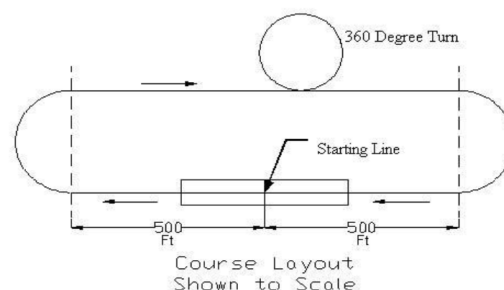


Figure 3.1: Competition Course Layout

3.1.1 Mission 1 – Staging Flight

There is no payload for M1. The aircraft must takeoff within 60 feet, and complete 3 laps of the course within a five-minute window. Landing may occur outside the window, but the aircraft must successfully land to get a score. M1 scoring is calculated as follows:

$$M1 = 1.0 \quad (3.1)$$

The maximum achievable score for Mission 1 is **1.0 point**, which is earned upon completion.

3.1.2 Mission 2 – Surveillance Flight

The payload for the Surveillance Flight is the Electronics Package. The loaded aircraft must take off within 60 feet. There is a ten-minute window for this mission, and teams fly as many laps as possible within the window. A lap is counted when the aircraft crosses the start/finish line. Landing may occur outside the window, but the aircraft must successfully land to get a score. The Electronics Package must also weigh at least 30% of the M2 AUW. M2 scoring is calculated as follows:

$$M2 = 1 + \frac{(\#laps * payload\ weight)_{WUDBF}}{(\#laps * payload\ weight)_{Max}} \quad (3.2)$$

where *#laps* is the number of successfully flown laps during the time window, and *payload weight* is the weight of the electronics package. The WUDBF and Max subscripts denote the *#laps*payload weight* achieved by WUDBF and the best *#laps*payload weight* achieved by any team respectively. The maximum M2 score is **2.0 points**, with one point coming from completion and up to one point awarded for performance.

3.1.3 Mission 3 – Jamming Flight

The payload for the Jamming Flight is the Jamming Antenna. The antenna is mounted to the side of the wing opposite the flight safety line. The aircraft must takeoff within 60 feet, and complete 3 timed laps of the course within a five-minute window. A lap is counted when the aircraft crosses the start/finish line. Landing may occur outside the window, but the aircraft must successfully land to get a score. Mission 3 scoring is calculated as follows:

$$M3 = 2 + \frac{\left(\frac{antenna\ length}{mission\ time}\right)_{WUDBF}}{\left(\frac{antenna\ length}{mission\ time}\right)_{Max}} \quad (3.3)$$

where *antenna length* is the length of the Jamming Antenna, and *mission time* is time it took the aircraft to complete three laps. The maximum M3 score is **3.0 points**, with two points coming from completion and up to one point awarded for performance.

3.1.4 Ground Mission – Structural Margin Demonstration

The Ground Mission is a demonstration of the aircraft's structural margin. There is a ten-minute window for this mission. The assembly crew member will install the heaviest payload configuration declared at tech inspection. The pilot will verify that all flight controls are working properly, then the assembly crew member will install the structural test fixture onto the aircraft's wing tips. The assembly crew member will apply test weights to the fuselage, inboard of the wing attachment points. They will continue to apply

weights until max weight is called or ten minutes passes. After the final test weight is applied, there will be a 30-second hold and the pilot will verify that the flight controls are still functioning. No structural failure or deformation must occur for the test to be successful. GM scoring is calculated as follows:

$$GM = \frac{\left(\frac{\text{test weight}}{\text{max weight}}\right)_{WUDBF}}{\left(\frac{\text{test weight}}{\text{max weight}}\right)_{Max}} \quad (3.4)$$

where *test weight* is the recorded weight of the aircraft with all test weights applied and *max weight* is the maximum weight of the aircraft declared at tech inspection. The maximum GM score is **1.0 point**, with one point being awarded for performance.

3.1.5 Total Competition Score

The Mission 1, Mission 2, Mission 3 and Ground Mission scores will be combined to give the Total Mission Score:

$$\text{Total Mission Score} = M1 + M2 + M3 + GM \quad (3.5)$$

The maximum achievable Total Mission Score is **7.0 points** with 4.0 points earned based on completion and 3.0 points earned based on performance. The Total Mission Score is then multiplied by the Design Report Score and added to the Participation Score to give the Total Competition Score (SCORE):

$$\text{SCORE} = \text{Design Report Score} \times \text{Total Mission Score} + \text{Participation Score} \quad (3.6)$$

3.1.6 General Mission Requirements

In addition to the requirements specific to each mission, there are also a number of other rules of note that apply to the competition aircraft during all missions:

- The aircraft and payloads must fit inside a shipping box with linear dimensions that sum to a max of 62 inches
- The maximum weight of all components in the shipping box must not exceed 50 lbs
- The minimum dimensions of the Electronics Package are 3.00 x 3.00 x 6.00 inches
- The Jamming Antennas must be a section of unmodified ½ inch Schedule 40 PVC pipe. Teams are allowed to bring three different lengths of pipe, but all must be carried in the shipping box.
- The antenna must be attached to the wingtip by means of an adapter, and must project vertically above the wing. The adapter must also be used to attach the aircraft to the ground fixture.
- The aircraft may utilize either NiCad/NiMH OR Lithium Polymer (LiPo) batteries for propulsion, but the total propulsion power total stored energy cannot exceed 100 watt-hours.
- The flight missions must be flown in order; a new mission cannot be flown until the team has obtained a successful score for the preceding mission.

3.2 Translation to Design Requirements

Based on the mission requirements outlined above, WUDBF formulated design requirements for each mission. These design requirements are broken down in the table below.

Table 3.1: Translation of Mission Requirements into Design Requirements

Mission	Mission Requirement	Design Requirement
All	60 ft TOFL	Low W/S, high $C_{L,max}$, high T/W
M2	Maximize laps	High cruise velocity, high T/W, low drag, high propulsive efficiency
	Maximize payload	Large wing area, sufficient internal volume
M3	Minimize lap time	High cruise velocity, high T/W, low drag
	Maximize antenna length	High stability, sufficient trim
GM	Maximize test weight	High strength of wing spars and wing attachment joints, small wingspan
	Minimize max flying weight	Light empty weight
General	Complete every mission	Low risk design, high stability
	Max propulsion energy of 100 watt-hours	High efficiency, constrains maximum propulsion power
	Must fit in shipping box	Constrains wingspan, overall length, tail dimensions and propeller diameter

3.2.1 Analysis of Design Requirements

TOFL: The takeoff limit imposes several design requirements. The aircraft must have a high $C_{L,max}$, a low wing loading, and a high thrust to weight. These elements are most critical for Mission 2, as that is where the heaviest payload is likely to be carried.

Mission 2: Mission 2 scoring is a function of the Electronics Package weight and the number of laps flown in ten minutes. These two requirements are contradictory, since a heavier payload requires a larger wing area or a more powerful motor to achieve the TOFL requirement. A larger wing area increases drag and decreases cruise velocity, which negatively affects the number of laps flown. A more powerful motor increases cruise speed but decreases endurance since the battery capacity is capped at 100 Wh. The payload weight should balance maximizing score with minimizing the size of the aircraft, and the propulsion system should balance high cruise velocity with enough endurance to best use the time interval.

Mission 3: Mission 3 scoring is a function of lap time and antenna weight. Since the aircraft must only fly three laps, endurance is not a concern as it was in Mission 2. Thus, the design must only balance antenna length with cruise speed, since a longer antenna increases drag. The antenna must also not be so long as to destabilize the aircraft, and overall stability should be strongly considered during the design process.

Ground Mission: The Ground Mission score is the maximum weight the aircraft can support by the wingtips divided by the maximum mission weight of the aircraft including payload. Since the Mission 2 payload will most likely be heavier than Mission 3, the Ground Mission is impacted by the Mission 2 configuration. The aircraft will be supported by the wingtips, so the wing spars and wing attachment joints should be as strong as possible. The aircraft's structural margin should also be balanced with its empty weight because a greater empty weight will negatively impact the GM score. The GM should also be considered when deciding the M2 payload, since it is possible that a lighter Electronics Package will overall increase the total mission score, even if it decreases the Mission 2 score.

Minimal Risk: Though mission performance is an important part of the competition score, over half of the mission points are awarded for completion. As such, the most important aspect of the aircraft's design is its overall stability and reliability.

Constraints: The 100 Wh battery limit constrains the propulsive power of the aircraft and so necessitates an efficient design. The shipping box constraint limits several dimensions of the aircraft. Since the wings are detachable, the limiting dimensions are likely to include the tail span and height, the propeller diameter, and the length of the aircraft. The 30% weight requirement for the Electronics Package sets a minimum AUV for that mission based on the aircraft's empty weight, which thus puts a lower bound on the aircraft's wing area since the aircraft must takeoff within the 60 ft limit.

3.3 Scoring Sensitivity Analysis

Based on the design requirements formulated above, WUDBF determined that there was no single obvious design direction. Prioritizing cruise velocity to maximize number of laps in M2 and minimize mission time in M3 drives the design towards a smaller, faster aircraft. Prioritizing M2 payload weight drives the design toward a larger, slower aircraft. Therefore, WUDBF conducted a scoring sensitivity analysis to determine which design direction maximizes total score.

3.3.1 Scoring Model

Given that each of the missions contain contradictory performance requirements, no obvious optimization path exists to achieve the maximum total mission score. Therefore, WUDBF developed a scoring model to analyze the tradeoffs between the different scoring variables. Two independent variables were identified: Jamming Antenna length and Electronics Package weight. The basic aircraft dimensions can be determined from the package weight, the desired takeoff distance, and wing loading. M2 and M3 cruise velocities can then be calculated, with the added antenna drag affecting the M3 cruise velocity. The GM score can be calculated from a known M2 payload and wingspan.

WUDBF then constructed a model to take those two independent variables as inputs. The model then linked these parameters to the rest of the scoring variables through a series of equations and calculated a

total mission score for each configuration. A diagram depicting the full scoring model is shown in Figure 3.2. In order to implement the model, a number of simplifying assumptions had to be made. These assumptions and their associated justifications are detailed in Table 3.2.

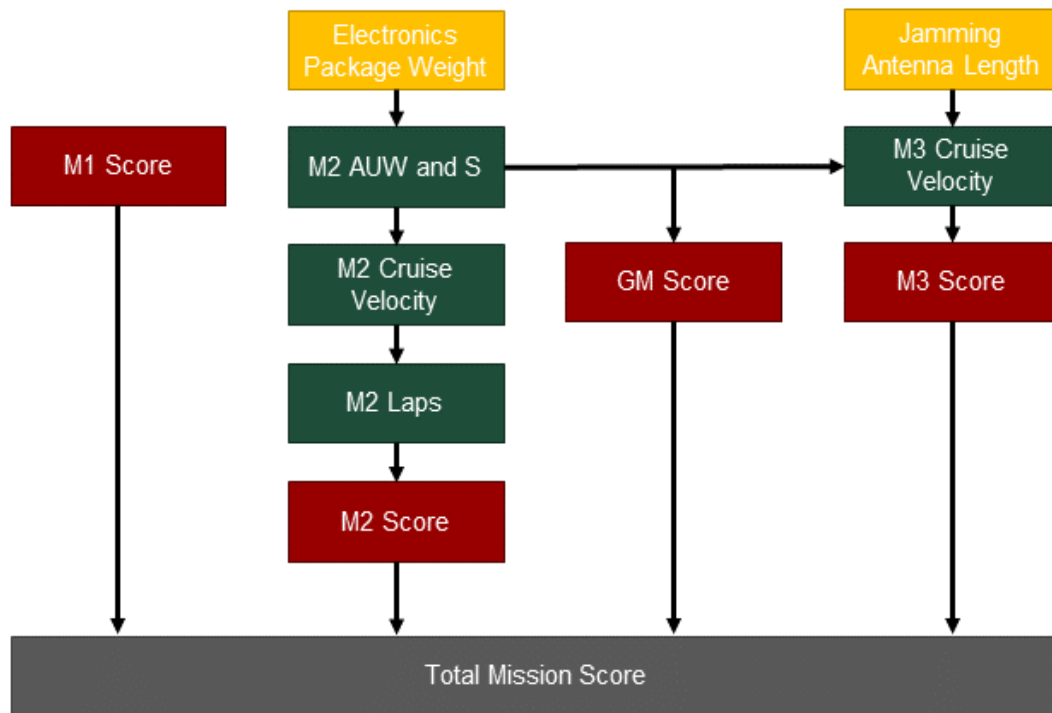


Figure 3.2: Scoring Model Flowchart

A sensitivity analysis was performed to determine the optimal combination of the Electronics Package mass and the Jamming Antenna length. Starting with reasonable estimates for aircraft weight and power, each input was varied. From each configuration, basic aircraft geometry was generated to predict flight performance. Configurations that did not meet the mission requirements were eliminated. M2 and M3 scores were calculated directly from the input variables, and the Ground Mission score was calculated using a three-point bending model to estimate the weight a aircraft with a given wingspan could support.

Table 3.2: Table of Key Assumptions and Justifications for Scoring Model

Assumption	Justification
Wing area varies between 0.1 and 0.5 m ²	An aircraft with a wing area less than 0.1 m ² is unlikely to take off with the required payload, and an aircraft with a wing area greater than 0.5 m ² is unlikely to fit in the shipping box.
Aspect ratio varies between 4 and 7	An aspect ratio below 4 is likely to be difficult to control and also reduces aircraft efficiency. Wings with an aspect ratio above 7 will likely be too large to fit in the box, complicating the application of the size constraint to the design space. [1]
The maximum antenna length is 1 m	A 1 m antenna approaches the predicted limit of the box constraint. Should the analysis show that a longer antenna is preferable, precise sizing can be performed once the box is constructed.
The maximum wing loading is 11 kg/m ²	Test pilot feedback on previous aircraft suggests a wing loading greater than 11 kg/m ² will make the aircraft very difficult to control due to a high stall speed. Greater wing loading will allow for a heavier M2 Payload and faster cruise velocity, but WUDBF prioritized safety by limiting the wing loading.
Max cruise velocity can be estimated from all-up weight (W), wing area (S), and antenna length.	Max cruise velocity can be determined by iteratively solving the following equation [2, p. 878]: $\rho S C_{Dmin} V_{max}^3 = \eta P + \sqrt{(\eta P)^2 - 4 W^2 V_{max}^2 C_{Dmin} k}$
Lap length is approximately 2500 feet.	Based on course diagram. 3.3.2

3.3.2 Results of Scoring Sensitivity Analysis

Running an analysis of over six million possible parameter combinations, the scores for each were calculated and normalized by the highest score in each mission to simulate the competition rules. The automatic points granted for completing each mission were not included to increase contrast between the mission scores. Contour lines of the M3 cruise velocity are also shown.

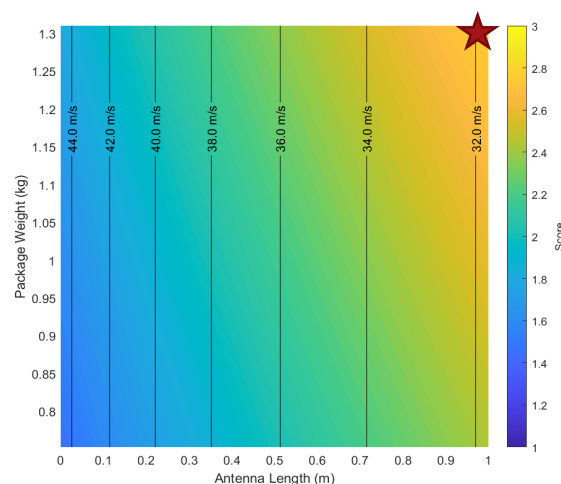


Figure 3.2: Plot of Scoring Variables and M3 Cruise Velocity

The input combination with the highest score is denoted with a star. Since the wing loading was held constant at 11 kg/m², the cruise velocity is only a function of the antenna length. The results indicate that antenna length has a greater impact on score than package weight. The optimal aircraft should be designed for a cruise velocity of around 32 m/s, carrying an Electronics Package of 1.3 kg. The length of the jamming antenna was limited to one meter, but the sensitivity plot suggests that increased score caused by a longer antenna outweighs the speed penalty caused by increased drag, so the aircraft should carry the longest antenna possible. Further stability testing will inform how long this antenna can be. The aircraft should be thus be optimized for stability to ensure a longer antenna can be carried. For this configuration, Mission 2 cruise velocity was calculated to be 44 m/s. A summary of preliminary design targets is shown below.

Table 3.3: Initial Design Targets

Design Specification	Initial Target Value
Electronics Package Weight	1.3 kg
Jamming Antenna Length	1 m
M3 Cruise Velocity	32 m/s
M2 Cruise Velocity	44 m/s
GM Test Weight	60 kg
M2 AUW	3 kg
M2 Thrust/Weight	1

These design targets were used to guide the initial conceptual design and were further refined in the preliminary design phase.

3.4 Configuration Selection

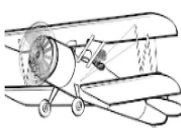
Once initial design targets were determined, decision matrices were then used to select the overall aircraft configuration as well as the configuration of major subsystems. All decision matrix weights were determined based on design goals and competition requirements. Individual criterion scores were then determined based on team knowledge and experience from previous competitions. The individual criterion scores were multiplied by the weights and summed together to determine the total weighted score for each configuration. A selection was then made by comparing the total weighted score.

3.4.1 Overall Aircraft Configuration

WUDBF first considered three overall aircraft configurations including the monoplane, biplane, and blended body configurations. Key design criteria included aerodynamic efficiency, stability and control, wing loading, and manufacturability. The highest weights were assigned to flight speed and stability and control as speed is important to earning points, and stability is a main driver of overall aircraft design. Using these criteria, the following configurations were compared in Table 3.4.

- **Monoplane:** Provides good aerodynamic efficiency, stability and control, and is likely the most feasible configuration. However, available wing area within the span limitation is the lowest of the considered configurations.
- **Biplane:** Potentially allows for more lift within the span limitation by maximizing available wing area. However, the increased drag and added structural weight associated with the configuration decreases efficiency.
- **Blended Body:** Allows for increased wing area while still maintaining relatively good aerodynamic efficiency. However, the complex geometry of the blended body reduces design and manufacturing feasibility. Additionally, the blended body configuration would have limitations in regard to stability and control.

Table 3.4: Aircraft Configuration Decision Matrix

		Rating (1-5)		
				
Criterion	Weight (%)	Monoplane	Biplane	Blended Body
Flight Speed	25	3	1	5
Drag	15	4	1	5
Wing Loading	15	4	5	3
Stability and Control	25	5	2	5
Manufacturability	20	5	4	1
Weighted Score		4.200	2.450	3.900
Rank		1	3	2

Ultimately, the monoplane configuration was selected due to its superior efficiency over the biplane, and easier manufacturability compared to the blended body. The blended body's extremely long shape may also have caused problems with respect to the box constraint.

3.4.2 Wing Placement

WUDBF next considered wing placement by comparing high-, mid-, and low-wing configurations. Stability and control and takeoff capability were weighted highest followed by feasibility, payload loading, and payload capacity. Using these criteria, the following configurations were compared in Table 3.5.

- **High-Wing:** Possesses the best stability and control due to its tendency to self-correct. Also is the most feasible configuration since WUDBF used it successfully in the past. However, access to the fuselage from above is restricted, increasing payload loading complexity.
- **Mid-Wing:** Has stability and control characteristics between that of the low- and high-wing configurations. However, this configuration necessitates that the structural support for the wing

runs through the center of the fuselage. This results in a decreased payload capacity and potentially increased payload loading complexity because the spar might conflict with either the battery or the payload mass.

- **Low-Wing:** Because the low-wing configuration allows for easy access from above, it has superior payload loading capability. However, is the least stable configuration due to the CG being above the wing.

Table 3.5: Wing Placement Decision Matrix

		Rating (1-5)		
				
Criterion	Weight (%)	High-Wing	Mid-Wing	Low-Wing
Maneuverability	15	3	4	5
Stability and Control	30	5	4	2
Takeoff Length	20	3	4	5
Manufacturability	15	5	2	3
Flight Speed	20	3	4	4
Weighted Score		3.900	3.700	3.600
Rank		1	2	3

Ultimately the high wing was selected because it was the most stable configuration.

3.4.3 Empennage Configuration

WUDBF then selected the empennage configuration out of the conventional tail, the T-tail, the H-tail, and the V-tail. The main design criteria for empennage configuration were determined to be stability and control, flutter, drag, wake interference during takeoff, and feasibility. Using these criteria, the following configurations were compared in Table 3.6.

- **Conventional:** Simple to design and manufacture. Also possesses good stability and control and flutter characteristics.
- **T-tail:** Keeps the HT in undisturbed air, which would reduce the drag. However, the high-mounted HT can also generate low flutter speeds and asymmetric lift and yaw which induces high torsional loads. Additionally, having the HT at the tip of the VT introduces additional complexity into the structure and manufacturing process.
- **H-tail:** Offers increased HT efficiency due to the endplates. However, the flutter speed of the H-tail is reduced due to the weight of the VTs being located at the tips of the HT. Additional control and structural complexity is introduced by having two VTs, decreasing feasibility.

- **V-Tail:** Possesses good flutter characteristics and reduces the wake interference at takeoff. However, the V-tail control system is significantly harder to design and implement.

Table 3.6: Empennage Configuration Decision Matrix

		Rating (1-5)			
					
Criterion	Weight (%)	Conventional	T-Tail	H-Tail	V-Tail
Drag	30	4	4	2	5
Stability and Control	40	3	4	5	4
Feasibility	30	5	3	2	2
Weighted Score		3.900	3.700	3.200	3.700
Rank		1	3	4	2

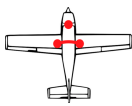
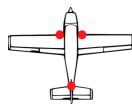
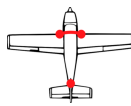
Ultimately, the conventional tail was chosen because it was considerably more feasible than the other configurations and provided improved stability/control and flutter characteristics with comparable wake interference.

3.4.4 Landing Gear Configuration

The landing gear was then chosen based on ground maneuverability, drag, and structural strength. Using these criteria, the following configurations were compared in Table 3.7.

- **Tricycle:** Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.
- **Bow Tricycle:** One front independent strut and two rear wheels which are connected to the fuselage via a common strut or fairing. This configuration allows good ground maneuverability while also increasing structural strength by adding bowed support to the two rear wheels. However, the structural strength of the front landing gear is a concern.
- **Strut Tail-dragger:** Two front wheels connected independently to the fuselage and one rear wheel connected to the tail. This configuration increases takeoff capability by allowing for an initial angle of attack during takeoff. However, it provides limited ground maneuverability and offers limited structural strength.
- **Bow Tail-dragger:** Two front wheels connected to the fuselage via a common strut or fairing and one rear wheel connected to the tail. Offers the same improved takeoff capability as the strut tail-dragger, while also providing better structural support. Ground maneuverability of this configuration is limited.

Table 3.7: Landing Gear Configuration Decision Matrix

		Rating (1-5)			
					
Criteria	Weight (%)	Tricycle	Bow Tricycle	Strut Taildragger	Bow Taildragger
Ground Maneuverability	30	5	5	2	2
Drag	35	4	4	4	3
Structural Strength	35	2	3	3	5
Feasibility	10	4	5	3	2
Takeoff Capability	10	5	5	3	3
Weighted Score		3.900	4.250	2.950	3.100
Rank		2	1	4	3

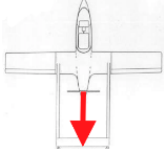
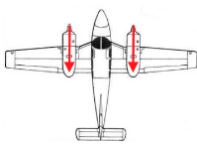
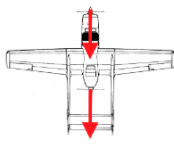
The Bow Tricycle configuration was chosen because it scored the highest by performing very well in ground maneuverability and feasibility compared to the other designs

3.4.5 Propulsion Configuration

The propulsion system was chosen based on its efficiency, thrust, weight, component interference, and drag. Using these criteria, the following configurations were compared in Table 3.8.

- **Tractor:** Single motor on the front of the aircraft. Provides the highest efficiency, lowest weight and drag, and is the most feasible. Ground clearance could be a concern depending on the required propeller size and height of front landing gear.
- **Pusher:** Single motor aft of the fuselage. Provides low weight and drag but has a decreased ground clearance (especially during rotation) and low efficiency due to fuselage interference.
- **Twin:** Two motors which are mounted on the wings. Provides improved ground clearance due to reduced propeller size and reasonable efficiency. Also offers torque balancing and is reasonably feasible. However, drag is increased resulting from the placement of the motors on the leading edge of the wing. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.
- **Push/Pull:** One motor at the front and aft of the fuselage. Offers low weight and drag. However, is the least efficient due to propwash interference, has the same ground clearance concerns as the tractor and pusher configurations, and is the least feasible. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.

Table 3.8: Propulsion Configuration Decision Matrix

		Rating (1-5)			
					
Criterion	Weight (%)	Tractor	Pusher	Twin	Push-Pull
Efficiency	35	5	4	3	3
Thrust	25	3	3	5	4
Weight	15	5	5	3	3
Component Interface	10	3	3	5	1
Drag	15	3	4	3	5
Weighted Score		4.000	3.800	3.700	3.350
Rank		1	2	3	4

The tractor configuration was ultimately chosen because it was thought to maximize efficiency while minimizing weight and drag. Additionally, the tractor configuration is the most feasible design and is one that WUDBF has successfully employed in the past.

3.5 Final Conceptual Design

The final conceptual design consists of a high-wing, monoplane, tricycle aircraft with a conventional tail and a propulsion system in the tractor configuration. The aircraft is designed to carry an M2 payload of 1.3 kg and a 1 m antenna. The targeted M2 and M3 cruise velocity is 44 and 32 m/s respectively. The targeted M2 AUW is 3 kg and M2 Thrust/Weight is 1.

4. Preliminary Design

During the preliminary design phase, the initial aircraft concept and configuration from the conceptual design phase was further developed. Key aspects of the design including the overall sizing, wing geometry, airfoil, and propulsion package were refined following a number of trade studies.

4.1 Design Methodology

The full design methodology employed by WUDBF is outlined in Figure 4.1 below.

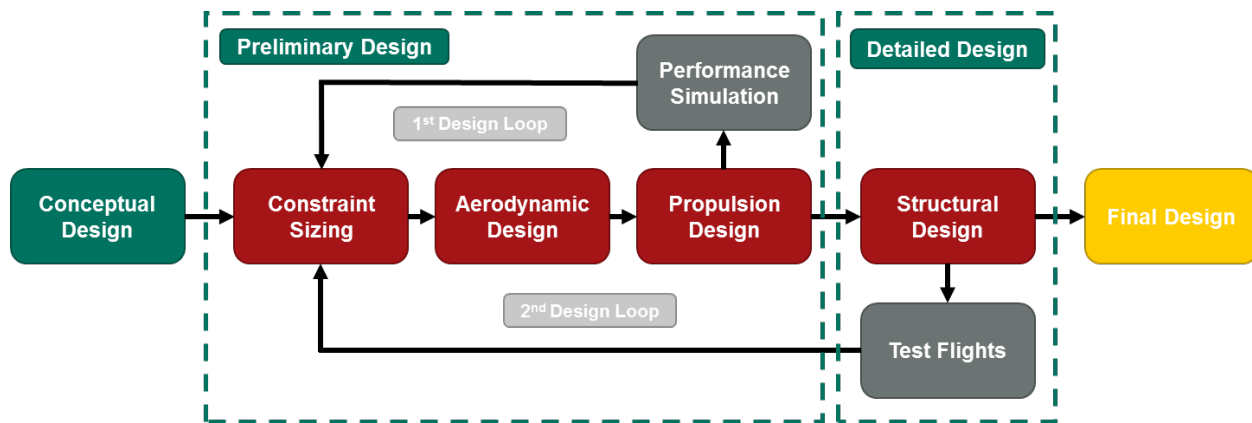


Figure 4.1: WUDBF Design Methodology

For the preliminary design phase, WUDBF utilized an iterative process focused on the performance of the fully loaded aircraft (i.e. with the Electronics Package). This was done because 1) the AUW for M2 is the greatest out of the flight missions, and therefore the constraints will be more limiting, and 2) aircraft are typically designed for the fully loaded case. Initial design targets from the conceptual design phase were used as a starting point. A trade study was performed on the wing geometry to determine the overall size of the aircraft. The chosen wing design was then used to guide the aerodynamic design, which dictated the airfoil and tail sizing. The performance and stability characteristics of the aircraft were analyzed using the vortex lattice method in the analysis tool XFLR. The drag polar of the aircraft was determined using induced drag data from XFLR and the component drag build-up method. Following the aerodynamic design, propulsion design took place using the target cruise, payload, drag polar, and endurance as starting points. The online tool eCalc was used to sort through a database of components to identify candidate propulsion system combinations. Data from eCalc was then used to select the optimal system, and performance was confirmed using WUDBF's time domain propulsion simulator written in MATLAB.

In order to iterate the design, WUDBF employed two design loops. Following the first design iteration, the preliminary design was tested using KM Sim, a kinematic mission simulator developed by WUDBF. Performance estimates from KM Sim were then used to iterate the preliminary design. Once the design met desired performance goals in KM Sim, the design moved into the second design loop. Structural design took place to create a detailed CAD model of the aircraft. This model was used to build a prototype aircraft. Flight test data from the prototype were then used to inform changes to the preliminary design. Ultimately, the final design developed as the culmination of several iterations, with each iteration improving on performance and mission scores.

4.2 Constraint Sizing

A trade study was conducted to determine the optimal wing geometry. Three constraints on the wing were identified: the shipping box size limit, the 60 ft takeoff length requirement, and a maximum wing loading of 11 kg/m² identified in section 3. Both the takeoff and wing loading constraints affect only the wing area,

but the wing loading constraint was determined to be the stricter constraint, and thus superseded the takeoff constraint in this analysis. For the box constraint, the limiting dimensions were determined to be the length of the aircraft, the horizontal stabilizer span, and the vertical stabilizer height. Since the wings are detachable and can be stowed alongside the aircraft, wingspan does not contribute to the limiting dimensions. These dimensions can all be estimated from wing area and aspect ratio using equations developed in Gundmusson [2]. Figure 4.2 shows these constraints plotted, as well as the predicted score of each combination.

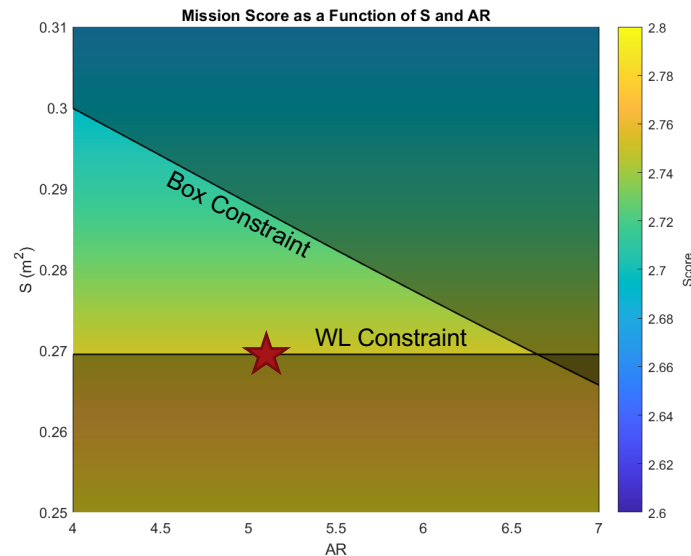


Figure 4.2: Plot Showing Required Wing Area and Aspect Ratio

The study showed that design space was extremely limited, with wing area ranging from 0.27 to about 0.295 m². Wing area had a greater effect on score than AR, since it more directly impacts cruise velocity. Aspect ratio was similarly limited, with the maximum being around 5.2. While maintaining the wing loading below 11 kg/m² was important for controllability, having a higher wing load would not disqualify WUDBF so long as the aircraft could takeoff within 60 ft. Exceeding the box constraint, on the other hand, would disqualify the aircraft, so a wing area of 0.27 m² and aspect ratio of 5.1 were selected to minimize aircraft drag while giving some margin for the aircraft to fit inside the shipping box.

4.3 Initial Weight Estimation

The AUW of the aircraft was estimated using a breakdown of individual components. The weight of the wing, HT, VT, and fuselage were estimated using previous aircraft, which were built with similar methods and materials. Battery, motor, servo, electronics, central CF rod, and landing gear weights were determined by weighing representative components already possessed by WUDBF. The M2 mass quantity was determined using an optimizer program.

Table 4.1: AUW Estimate Breakdown

Component	Weight (kg)
Wing	0.130
HT	0.033
VT	0.008
Fuselage	0.285
Battery	0.642
Motor	0.160
Servos	0.082
Electronics	0.150
Front Landing Gear	0.040
Rear Landing Gear	0.090
ESC	0.080
M2 Payload	1.300
Total M1:	1.700
Total M2:	3.000

Based on this analysis, the predicted AUW of the fully loaded aircraft during M1 is 1.7 kg, and during M2 it is 3.0 kg.

4.4 Aerodynamic Performance

The aerodynamic design of the aircraft was iteratively improved to ensure that it is stable, controllable, and achieves maximum aerodynamic performance. XFLR5 and MATLAB were the primary tools used for aircraft aerodynamic analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were dimensioned to improve handling characteristics and minimize takeoff distance, respectively.

4.4.1 Main Airfoil Selection

In order to select the main airfoil, the predicted Re of the aircraft was first calculated, using Equation 4.1:

$$Re = \frac{cV_{\infty}}{\nu} \quad (4.1)$$

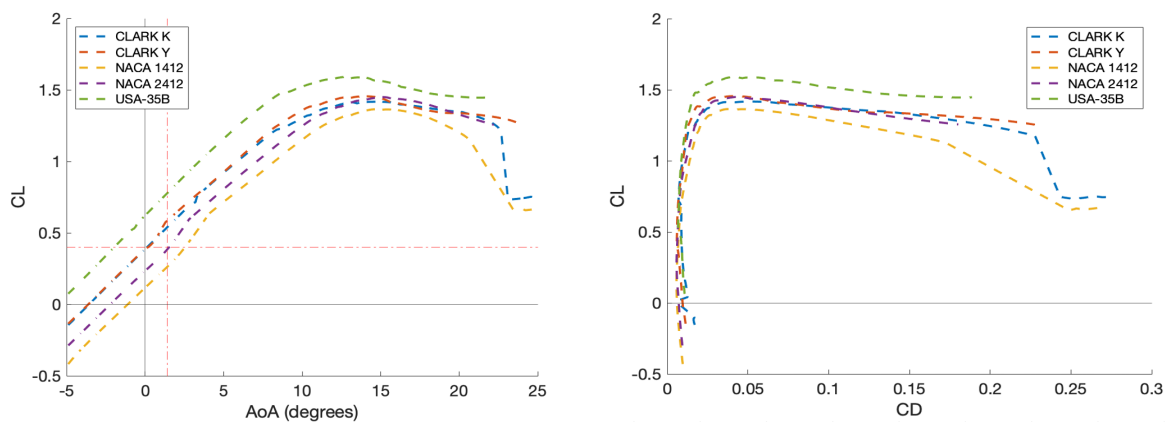
where c is the mean aerodynamic chord [m], V_{∞} is the freestream velocity [m/s], and ν is the kinematic viscosity of air [m²/s]. Kinematic viscosity was calculated from the average temperature in April (23 °C) and altitude (330 m) in Tucson which yielded a kinematic viscosity of 1.498×10^{-5} m²/s. These conditions yielded a Reynold's number of 450,384.

Airfoil selection was then performed by selecting several different airfoils with maximum C_L/C_D near a Reynold's number of 450,384. Additionally, the search was restricted to airfoils with a minimum thickness of 10% and a maximum camber of 8% to ensure that the airfoil could accommodate structural and system integration and be reasonably manufactured with the processes available. Five promising airfoils were selected for comparison as shown in the table below:

Table 4.2: Airfoil Selection Options [3]

	Max. C_L/C_D at $Re = 500\,000$	Max. Thickness [% of chord]	Max. Camber [% of chord]
Clark K	102.7	11.7	3.3
Clark Y	98.7	11.7	3.4
NACA 1412	73.5	12	1
NACA 2412	87.3	12	2
USA-35B	106.0	11.6	4.1

Once these potential airfoils were identified, analysis of each airfoil was performed in XFOil and XFLR to determine which provided a favorable lift coefficient, lift-to-drag ratio, and drag bucket at a Reynold's number of 450,384.

**Figure 4.3: Airfoil C_L vs AoA (left) and C_L vs C_D (right)**

As shown in the Figure above, the USA-35B possessed the highest lift coefficients and widest drag bucket. While the lift generated from this airfoil would improve T-O performance, it would also produce excessive lift at the target cruise velocity of 44 m/s, meaning the aircraft would have to fly slower resulting in longer lap times. Similarly, the Clark Y airfoil would also generate more lift than necessary at the target cruise velocity. While the NACA 2412 did not have the highest lift coefficients, it was found to provide sufficient lift at T-O and appropriate lift to meet cruise requirements. The NACA 2412 also had a drag bucket which comparable to the Clark Y. Thus, the NACA 2412 airfoil was chosen because it had the lowest C_{Dmin} , a wide drag bucket, and the desired lift characteristics at the target cruise velocity.

4.4.2 Empennage Sizing

Stabilizer airfoils are generally symmetric as they may either provide upforce or downforce, depending on the final trim state of the aircraft. WUDBF selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects. Initial sizing of the empennage for stable flight began by

investigating the vertical tail (VT) and horizontal tail (HT) volumes for aircraft of the same class. For a single-engine aircraft, the initial estimate was set at around $V_{HT} = 0.60$ and $V_{VT} = 0.04$ [4]. Initial horizontal and vertical tail areas were then solved for using Equation 4.2 and 4.3 as shown below [2, p. 513]:

$$S_{HT} = \frac{V_{HT} * S_{ref} * c_{ref}}{I_T} \quad (4.2)$$

$$S_{VT} = \frac{V_{VT} * S_{ref} * b_{ref}}{I_T} \quad (4.3)$$

Where S_{HT} is the horizontal tail area [cm^2], V_{HT} is the horizontal tail volume [cm^3], S_{VT} is the vertical tail area [cm^2], V_{VT} is the vertical tail volume (in cm^3), S_{ref} is the main wing area [cm^2], c_{ref} is the mean geometric chord [cm], b_{ref} is the span of the main wing [cm], and I_T is the optimal tail arm length [cm].

Based on these calculations, the VT and HT planform areas were set at 353.2 cm^2 and 767.7 cm^2 respectively. An initial assumption was made that the AR of the HT would be equal to about half the AR of the main wing and that the AR of the VT would be about 2 [5]. Initial estimates for the span of the VT and HT were then made using Eq 4.4 and 4.5 as shown below [2, p. 513]:

$$b_{HT} = \sqrt{AR_{HT} * S_{HT}} \quad (4.4)$$

$$b_{VT} = \sqrt{AR_{VT} * S_{VT}} \quad (4.5)$$

Where b_{HT} is the horizontal tail span (in cm), and b_{VT} is the vertical tail span (in cm).

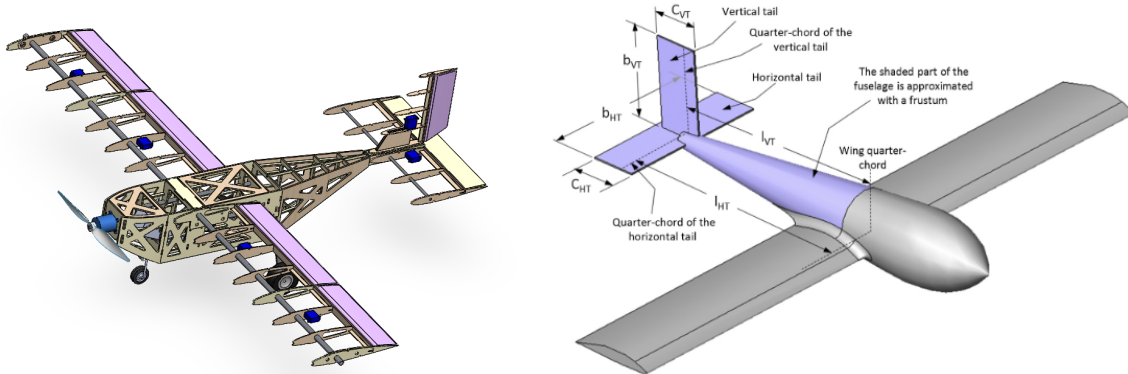


Figure 4.4: Initial Configuration of the *Bear Force One* (left) vs. Simplified Aircraft Layout used by Gudmundsson (right) [2].

After sizing the rudder and vertical stabilizer, we verified that it would be able to turn properly, even with the antenna. To do this we reserved half of the moment generated by the rudder and compared it to the moment generated by the antenna about the CG. The moment generated by the rudder times 1/2 would have to be greater than the moment of the antenna. This analysis showed the aircraft would be able to turn with any antenna height below 1.65 m. This is shown in the figure to the right

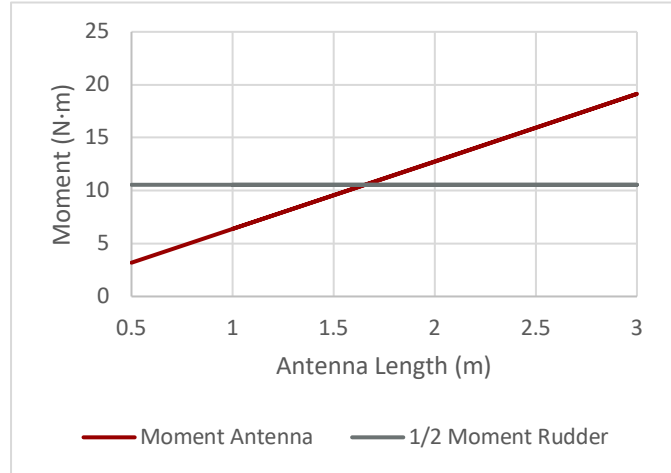


Figure 4.5: Turning Capability vs Antenna Height

4.4.3 High Lift Devices

Since we were adding an antennae to the end of the wing in M3, it was decided that there was no room for high lift devices on the ends of the wing, as they would be interfered with by the antenna, require a complicated structure to maintain, and they would add to the length of the wings, making it harder to pass the box size constraint. The cost in weight and drag that high lift devices would add would outweigh the benefit in additional lift that they would give the aircraft.

4.4.4 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using rule-of-thumb parameters from Gudmundsson [2, p. 419]. Control surface sizing was verified using Equation 4.6 (see below) from Gudmundsson [2, p. 953], previous WUDBF aircraft, and prototype performance.

$$C_{l_{\delta a}} = \frac{c_{l_{\delta a}} C_R}{Sb} \left[(b_2^2 - b_1^2) + \frac{4(\lambda-1)}{3b} (b_2^3 - b_1^3) \right] \quad (4.6)$$

Where $C_{l_{\delta a}}$ is the aileron authority, $c_{l_{\delta a}}$ is the change in lift coefficient with aileron deflection, b is the wing span [m], S is the wing area [m²], C_R is the root chord [m], and λ is the taper ratio.

Plain flaps were chosen to simplify the manufacturing process. The hinge point of the flaps was placed at the center of the leading edge arc to minimize large gaps and overhangs when the surfaces are deployed. The flaps were given a chord fraction of 30%, and a span of 40% of the wing.

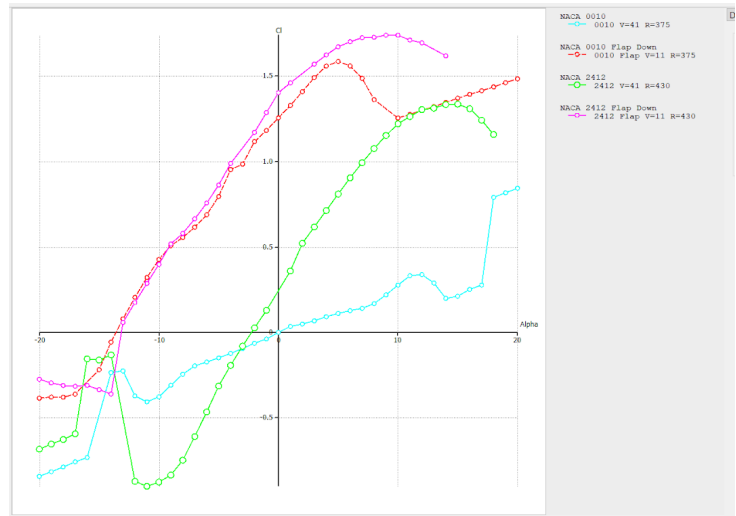


Figure 4.6: C_L vs AoA With and Without Flaps

The ailerons had a chord fraction of 30% and a span of 60% of the wing. Both the flaps and the ailerons were designed to deploy between -20 and 20 degrees. Using data taken from XFLR, Figure 4.5 shows the plotted lift coefficient versus AoA with the flaps deployed at 20° and with no flaps. Chord fractions for the control surfaces on the empennage were 30% for the elevator, and 40% for the rudder. The rudder was designed to deflect a maximum of 20°, while the elevator was designed to deflect a maximum of 20°.

4.4.5 Dynamic Stability Analysis

The dynamic stability characteristics of the aircraft were analyzed using XFLR. The Root Locus plot below in Fig 4.7 shows the longitudinal dynamic stability modes of the aircraft in M1 and M2.

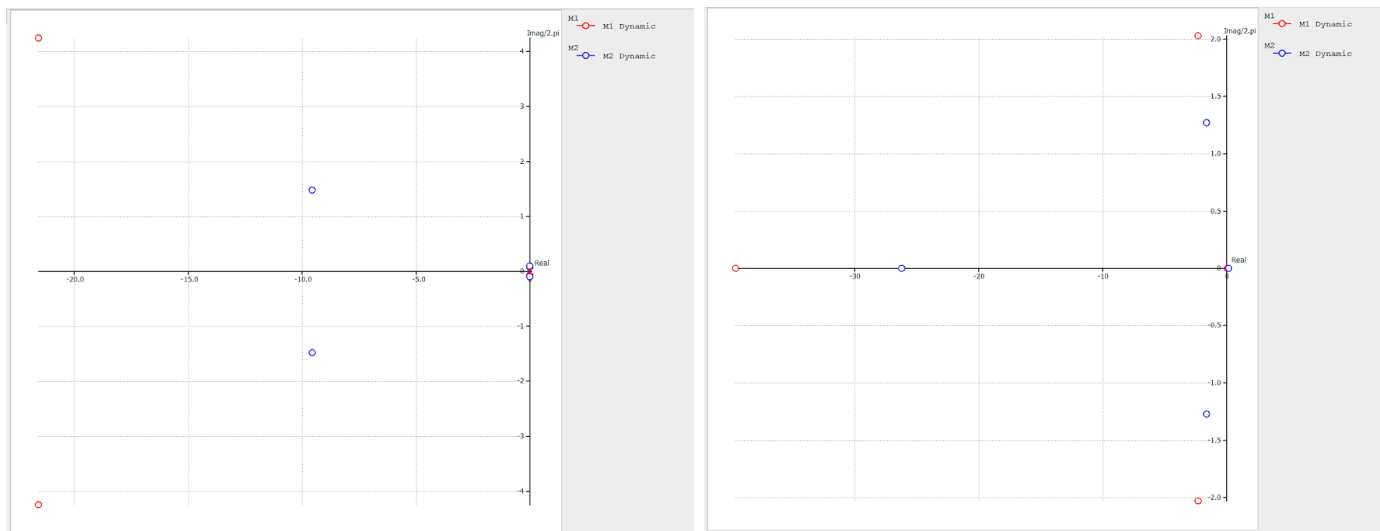


Figure 4.7: Root Locus Longitudinal View (left) and Lateral View (right)

Based on the root locus plots, all modes were stable for both loading conditions. The design goal of this aircraft was to satisfy the level 1 flying qualities of a Class 1 aircraft described in MIL-F-8785C [12]. The dynamic stability characteristics from XFLR with the aircraft in M1 and M3 configurations are shown below in Table 4.3 with MIL-F-8785C requirements in parenthesis. Similarly, the stability characteristics of the aircraft in the M2 configuration are shown below in Table 4.4.

Table 4.3: Dynamic Stability Table for M1 and M3

Mode	ζ	ω_n [rad/s]	$\zeta\omega_n$ [rad/s]	τ [s]
Short Period	0.630 ($0.35 < \zeta < 2.00$)	5.448	4.233	-
Phugoid	0.006 ($\zeta > 0.004$)	0.069	0.69	-
Roll	-	-	-	0.023 ($\tau < 1.4$)
Dutch Roll	0.176 ($\zeta > 0.19$)	2.056 (> 1.0)	2.024 (> 0.35)	-
Spiral	-	-	-	19.709

Table 4.4: Dynamic Stability Table for M2

Mode	ζ	ω_n [rad/s]	$\zeta\omega_n$ [rad/s]	τ [s]
Short Period	0.473 (0.35 < ζ < 2.00)	2.118	1.475	-
Phugoid	0.014 (ζ > 0.004)	0.095	0.095	-
Roll	-	-	-	0.035 (τ < 1.4)
Dutch Roll	0.198 (ζ > 0.19)	1.282 (> 1.0)	1.6173 (> 0.35)	-
Spiral	-	-	-	4.563

Thus, the dynamic stability characteristics of the M2 loaded configuration are slightly less stable than M1 longitudinally; however, it is mostly comparable to M1, and it is in the stable range. The values for the damping ratios of the various movements is generally within the stability parameters we assigned to an aircraft of this type based on the damping ratios of similar aircrafts. The Dutch roll stability was slightly below the desired value but was deemed acceptable from pilot feedback.

4.4.6 Drag Analysis

Total aircraft drag was calculated using the component drag build-up method (CDBM) as described by Gudmundsson [2]. The method models the total drag C_D as the sum of the minimum drag C_{Dmin} and the lift induced drag $C_{D,i}$. The minimum drag coefficient, C_{Dmin} , AKA the zero-lift or parasitic drag coefficient was calculated using the following expression:

$$C_{Dmin} = \left(\frac{1}{S_{ref}} \sum_{i=1}^N C_{f_i} \times FF_i \times IF_i \times S_{wet_i} + C_{Dmisc} \right) \times \left(1 + \frac{CRUD}{100} \right) \quad (4.7)$$

Where S_{ref} is the reference area (in cm²), C_f is the skin friction coefficient, FF is the form factor, IF is the interference factor, S_{wet} is the wetted area of the component (in cm²), and C_{Dmisc} is the drag coefficient from miscellaneous components other than the fuselage, HT, VT, and main wing.

The skin friction coefficient for the main wing was determined using Young's method for mixed laminar-turbulent flow [2, p. 679]. The FF for the main wing was found using Raymer's method with the compressibility correction term set equal to one [4]. Finally, the IF was found by using a typical IF for an un-filletted low wing. The skin friction coefficients for the HT and VT were also determined using Young's method for mixed laminar-turbulent flow, and the FF s were found using Jenkinson's method [2, p. 701]. A typical IF for a conventional tail was used. The skin friction coefficient for the fuselage was determined by considering a 100% laminar boundary layer. The FF of the fuselage was then estimated using the method suggested by Hoerner [6, p. 6-17] for bodies in airflow $Re > 105$. Air inlet vents and winglet endplates were accounted for in the C_{Dmisc} term using equivalent parasitic coefficients. To ensure that the drag estimate remained conservative, drag reduction from winglet endplates was ignored. To include drag contributions that could not otherwise be accounted for, a CRUD factor of 20% was applied as recommended by Gudmundsson (see Equation 4.7). From these calculations, the minimum drag

coefficient for missions 1 and 2 was found to be 0.056. The full parasitic drag breakdown is presented below:

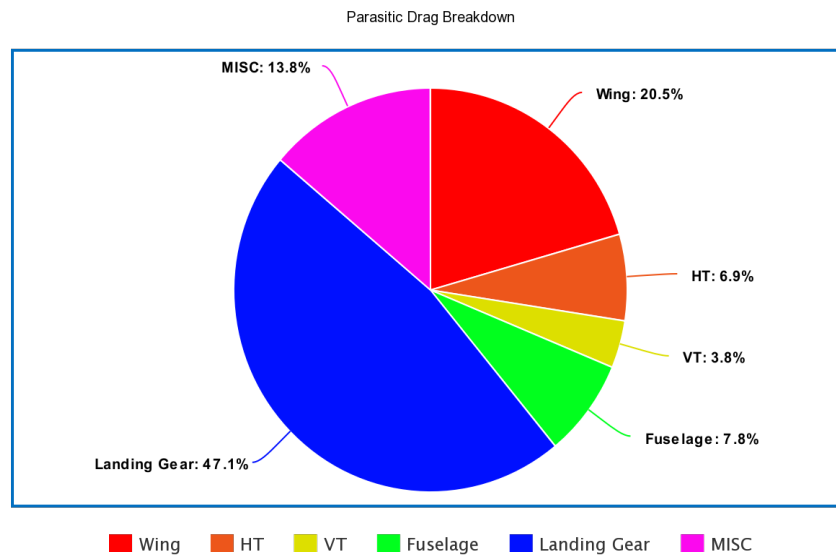


Figure 4.8: Parasitic Drag Breakdown M1 and M2

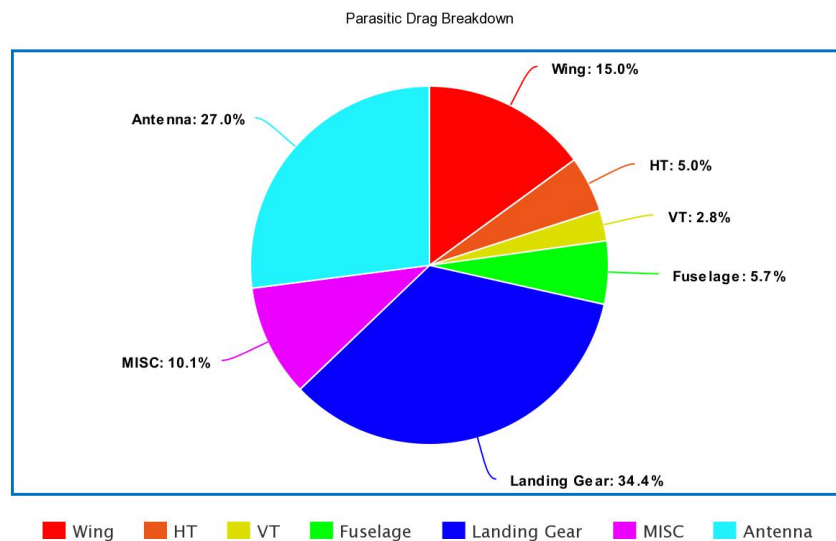


Figure 4.9: Parasitic Drag Breakdown M3

The landing gear was the largest contributor to parasitic drag across all mission configurations. For the M3 configuration, the antenna contributed a significant amount of drag, which will lower the cruise velocity as predicted in Section 3.3. The parasitic drag was then added to the lift induced drag, calculated using XFLR, to give the total drag polar, as shown in the figure below:

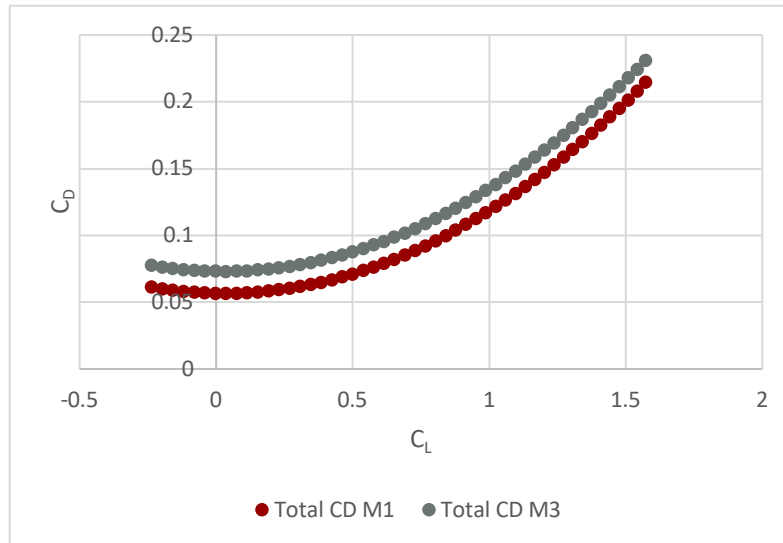


Figure 4.10: Total C_D vs. C_L

4.5 Propulsion

The preliminary design of the propulsion system focused on meeting multiple requirements set earlier in the design process using M2 as the limiting case. In particular, a required thrust of ~ 3 kgf, a desired cruise velocity of ~ 44 m/s, and a desired flight time of 10 minutes. M2 was chosen as the limiting case because of the increased payload weight and the high endurance needed to complete the maximum number of laps. Additional consideration included maximizing efficiency and minimizing propulsion overall system weight.

4.5.1 Battery Selection

The DBF rules allow for the use of either LiPo (Lithium Polymer) or NiMH (Nickel-Metal-Hydride) batteries for propulsion. While NiMH batteries tend to be more robust and safe, LiPo batteries offer higher energy density, discharge rate, efficiency, and reliability. Therefore, WUDBF chose to use LiPo batteries for the propulsion pack.

LiPo's are commonly available in 2s, 3s, 4s, 5s, 6s, and 8s packs. For the desired cruise velocity, LiPos below 6s are unlikely to provide the performance required; therefore, 6s and 8s LiPos were chosen as the main considerations. Additionally, the 100 Wh constraint on the propulsion power limited the battery capacity available for each battery. For 8s LiPos, the highest commercially available capacity under 100 Wh is 3300 mAh (94.7 Wh). The lower capacity of the 8s LiPos greatly decreases the flight time the aircraft would be able to achieve as compared to a 6s LiPo, which have a capacity of 4500 mAh commercially available. Furthermore, 6s LiPos with 4500 mAh (99.9 Wh) are more readily available than 8s LiPos and take the most advantage of the 100 Wh limit. The Admiral 4500mAh 6S 45C LiPo battery was selected to serve as the propulsion pack since it met the capacity requirement.

4.5.2 Motor and Propeller Selection

In order to identify possible motor-propeller combinations, WUDBF employed the online propulsion design tool eCalc [7]. Using the parameters listed in the table below, WUDBF searched through the eCalc database for motor-propeller combinations that met the stated performance requirements.

Table 4.5: eCalc Search Parameters

Parameter	Value	Reasoning
T/W	~1	See Section 3.3
Max Cruise Velocity [m/s]	44	See Section 3.3
Endurance [min]	10	See Section 4.5.1
Battery	6s, 4500 mAh	See Section 4.5.1
Max Prop Diameter [in]	12	Estimated max ground clearance

For this search, a list of nearly 2000 unique motor-propeller combinations were identified. We limited the search to only trusted manufacturers, such as SunnySky and Scorpion, that were available, and in-stock, at the time of the search. This created a new list of about 10 motor-propeller combinations. The list was then further refined to most closely match the previously stated requirements, narrowing the possibilities down to 4 unique motor-propeller combinations shown in Table 4.7:

Table 4.6: Possible Motor-Propeller Combinations

Motor	Prop	Elec. Power @ Max [W]	Current @ Max [A]	RPM (Static)	Speed [m/s]	Static Thrust [kgf]	T/W	Mixed Flight Time [min]	Drive Weight [kg]
SunnySky X3120-585 V3	12x8	980.5	46.89	10534	34.2	4.128	1.38	9.7	1.010
Scorpion A-3017-600	12x7	916.3	43.64	10761	31.9	3.769	1.26	9.8	0.985
Scorpion SII-4020-630	11x7	863.9	40.99	11898	33.0	3.549	1.18	10	1.152
SunnySky X4120-480 V3	12x8	693.6	32.56	9527	30.8	3.376	1.13	12.2	1.156

All four choices met or nearly met the stated required endurance of 10 minutes. However, all motors failed to meet the speed requirement of 44 m/s. The sensitivity analysis did not account for decreasing propulsive efficiencies at higher velocities, explaining this discrepancy. The SunnySky X3120-585 V3 and Scorpion SII-4020-630 had the highest listed cruise velocities, and so were given further consideration. Thus, in order to further evaluate each combination, plots of endurance vs throttle, efficiency vs throttle, speed vs throttle, and thrust vs electrical power were generated using data from eCalc:

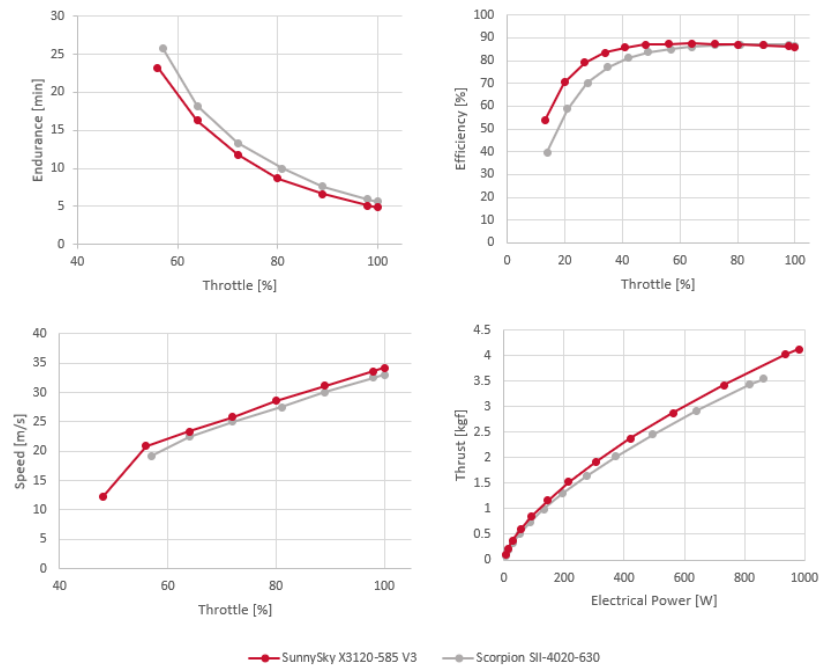


Figure 4.11: Performance Data for Motor Propeller Combinations

From Figure 4.10, the SunnySky X3120-585 V3 motor had greater or equal electrical efficiency, speed, and thrust than the Scorpion SII-4020-630. The endurance of the SunnySky, while overall lower than the Scorpion, very nearly meets the required time of ten minutes (as can be seen in eCalc's calculation of "mixed flight time" in Table 4.7). Additionally, the SunnySky has a lower drive weight than the Scorpion, as can be seen in Table 4.7, while providing superior performance over the Scorpion. Based on these factors, the SunnySky X3120-585 V3 was ultimately chosen for the aircraft's propulsion system.

4.5.3 ESC Selection

Given the predicted max current of 41.02 for the chosen propulsion system, WUDBF chose to use the Phoenix Edge Lite 75 A (8S) as the ESC. The Phoenix Edge Lite can handle 75 A continuous current, which provides an adequate safe margin. The selected propulsion pack had a maximum continuous discharge current rating of 180 A, which is over the given 100 A limit for an arming fuse. Thus, WUDBF went with a 100 A blade style automotive fuse for the aircraft.

4.5.4 Refinement for M3

Given that the propulsion system was designed primarily with M2 in mind, WUDBF next considered whether it could be refined to improve M3 performance (since the competition rules allow for the propeller and battery to be swapped out for different missions).

WUDBF first considered whether the propeller could be optimised to increase cruise velocity during M3. In general, increasing the propeller pitch increases pitch speed, which increases efficiency at higher velocities and allows for higher cruise velocity. This comes with the sacrifice of efficiency at low

airspeeds, negatively impacting static thrust and takeoff performance [4]. However, increasing the pitch of the propeller any further leads to the blade stalling and decreasing the static thrust below acceptable levels. Thus, the 12x8 propeller was selected for use in both M2 and M3.

WUDBF then considered changing the propulsion battery pack to increase the cruise velocity. By increasing all-up-weight, the wing loading increases and, therefore, the speed increases. However, since the battery selected is already the highest capacity and cell count that meet the 100 Wh limitation, adding any significant weight through the battery was not possible. Another method to achieve a higher cruise velocity is by selecting a battery with a higher C-rating, but at the time of search there were no available batteries with a higher C-rating than the selected battery pack. So, the Admiral 4500 mAh 6S 40C was kept for all three missions.

4.5.5 Required vs. Available Thrust Analysis

The dynamic performance of the propulsion system was evaluated to verify eCalc's predicted performance and to determine the aircraft's speed envelope. eCalc does not output dynamic thrust data, so this data was instead obtained from the APC Propellers website (the chosen propeller manufacturer) [8]. The APC performance data was generated based on vertex theory using actual propeller geometry and the NASA Transonic Airfoil Analysis Computer Program to estimate section lift and drag. Predicted RPM values from eCalc were used to query the APC dataset at velocities up to 50 m/s, with the corresponding dynamic thrust values calculated using an interpolation script written in MATLAB. The resulting dynamic thrust curve was then plotted along with the required thrust curve as shown below.

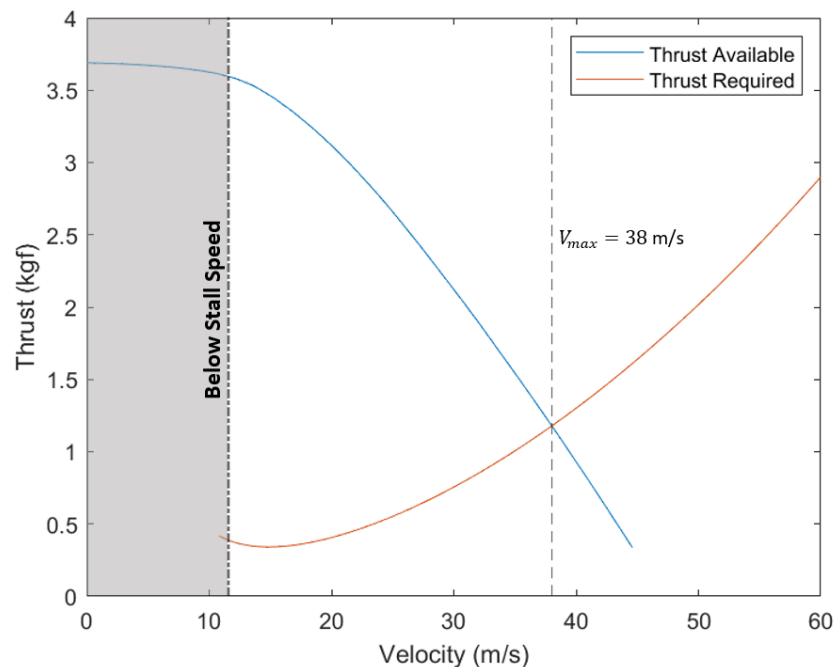


Figure 4.12: Required vs. Available Thrust Curves For Fully Loaded Aircraft

Based on this analysis, the estimated max cruise velocity for the fully loaded aircraft (in M3) is 38 m/s. The velocity calculated above varies from the value given by eCalc because propCalc (the calculator used to acquire the values from Table 4.7) uses multiple simplifications and assumptions for the drag on the aircraft, which may lead to a difference in the calculated speed vs the actual speed. The analysis done above more accurately takes into account the drag on the aircraft and thus has a more accurate estimate for the cruise speed of the aircraft.

4.6 Predicted Aircraft Performance

Km_sim, a kinematic mission simulator designed in MATLAB by WashU students, was utilized to test the feasibility of the *Bear Force One* design and simulate how the aircraft would perform in competition [9].

4.6.1 Brief Explanation of Km_sim

Km_sim works by simulating an aircraft's performance as it flies through a mission profile. Each mission profile consists of a series of functions based on simple kinematics and Newton's 2nd Law describing the motion of the aircraft during a particular phase of flight (e.g., takeoff, climb, cruise, etc.). Specific mission profiles were written for M1, M2, and M3. Aerodynamic and propulsion data from Sections 4.4 and 4.5, respectively, were used to generate aerodynamic and propulsion performance look-up tables. These tables allow the simulation to determine the lift, drag, and thrust of the aircraft at any particular time point. Km_sim steps through each mission phase, numerically integrating the relevant equations of motion using ode45 and the look-up tables, while keeping track of changes in the state variables. At the end of the simulation, plots are generated displaying how the state variables change over time as shown in the figures below. For a complete explanation of Km_sim and the source code see [9].

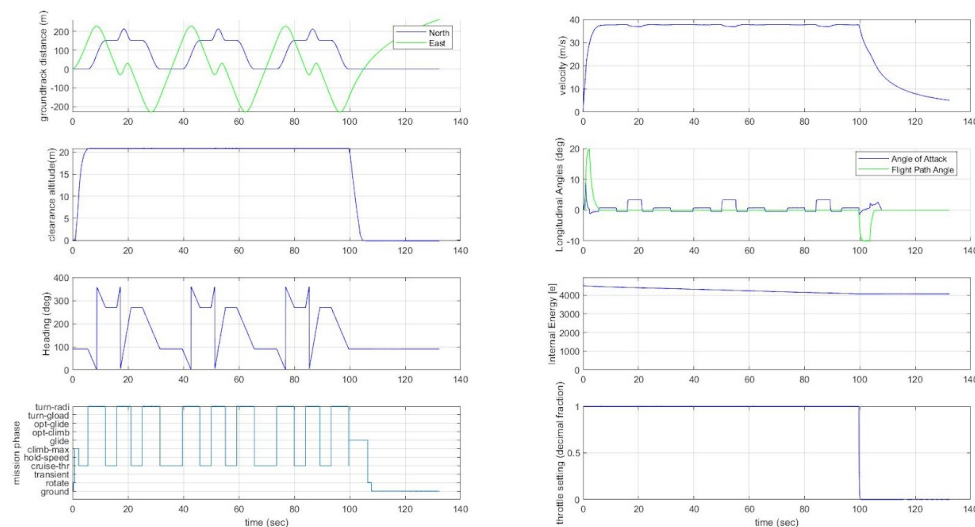


Figure 4.13: Plots of Different State Variable Outputs Versus Time During a Simulation of M1

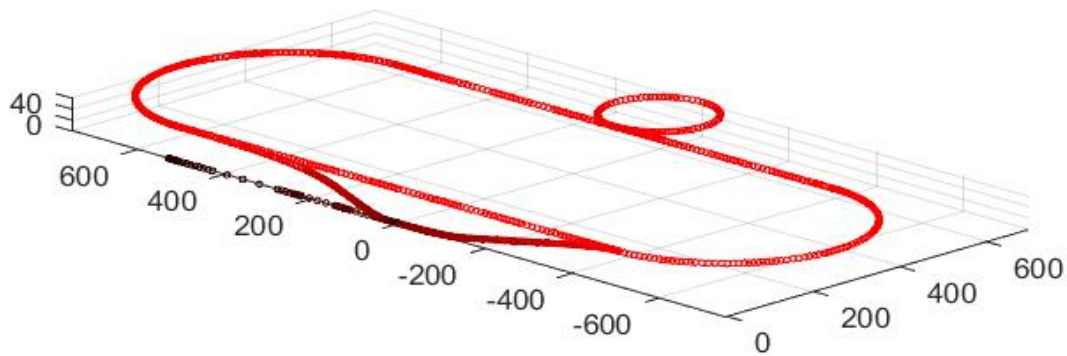


Figure 4.13: Trajectory Plot Illustrating the Path Taken by the Aircraft in 3D Coordinates During a Simulation of M1

4.6.2 Km_sim Results

Separate simulations for M1, M2, and M3 were carried out in Km_sim to estimate performance in competition. Normalized scores were calculated based on the maximum competition scores predicted as part of the sensitivity analysis in Section 3.3. Having a cruise throttle below 1 was considered for Mission 2 due to it being an endurance test. However, simulations indicated that a flight could last the full 10 minutes at full cruise throttle, allowing for optimal speed. Thus, the aircraft flies at full cruise throttle in every mission.

Table 4.7: Predicted Aircraft Performance

	Mission 1	Mission 2	Mission 3
Inputs			
All-Up Weight [kg]	1.70	3.00	2.11
Payload Weight [kg]	-	1.30	0.41
Wing Loading [kg/m ²]	6.22	10.97	7.72
Antenna Length [m]	-	-	0.86
C_{Lmax}	1.735	1.735	1.735
Parasitic Drag	0.05611	0.05611	0.13676
Cruise Throttle	1	1	1
Turn Load Factor	2.5	2.5	2.5
Outputs			
Takeoff Distance [ft]	17.16	32.45	22.31
$C_{Lcruise}$	0.18338	0.37875	0.18338
$C_{Dcruise}$	0.00214	0.00905	0.0024
Cruise Velocity [m/s]	37.78	37.70	30.76
Number of Laps	3	17	3
Mission Time at End of Final Lap [min]	1.75	9.82	2.19
Scoring			
Bear Force One Score	1	216.73	39.23
Normalized Score	1	1.49	2.42

4.7 Uncertainties

The design tools and analysis methods used throughout the preliminary design process all possessed a certain level of uncertainty. The lift and induced drag of the aircraft was computed in XFLR using the vortex lattice method (VLM). Using VLM, lifting surface thickness and air viscosity were neglected. Additionally, the fuselage was excluded from the analysis. Based on previous WUDBF designs, however, it was determined that the decreased lift due to neglecting viscosity is approximately equal to the lift gained by excluding the fuselage. It has also been verified that the contribution of drag associated with wing thickness was small compared to total drag calculations. Therefore, the lift and induced drag of the actual aircraft was expected to be approximately equal to the XFLR predicted lift using the VLM. The parasitic drag estimate using Gudmundsson's method were also subject to a certain level of uncertainty, but that uncertainty was mostly accounted for by the 20% CRUD factor.

eCalc and the APC propeller performance database were the two primary tools used for propulsion design. These third-party tools both carry a certain level of uncertainty owing to the fact that it is not entirely clear how they calculate their outputs. However, this uncertainty is mitigated by the fact that this method of propulsion design has been used successfully by both WUDBF and other DBF teams in the past with an acceptable level of accuracy.

Finally, Km_sim, while a very useful tool, makes a number of assumptions in its performance calculations. For instance, the simulation assumes that the pilot flies the aircraft on a perfect trajectory. The effects of wind are also neglected in the simulation. Since Km_sim relied on data from the aerodynamic and propulsion designs, any uncertainties from those processes were carried over into the simulation.

Ultimately, the nature of the iterative process meant that the uncertainty associated with any particular design tool was mitigated by ensuring that every aspect of the design agreed with the rest.

5. Detail Design

Following completion of the preliminary design, the structure and external geometry of the aircraft were finalized during the detail design stage (shown in Figure 5.1). The antenna and ground fixture adapter developed during this process as well as the housing for electronics and the M2 payload. Throughout, structural strength, reliability, weight, and mission performance were considered to refine the competition aircraft.

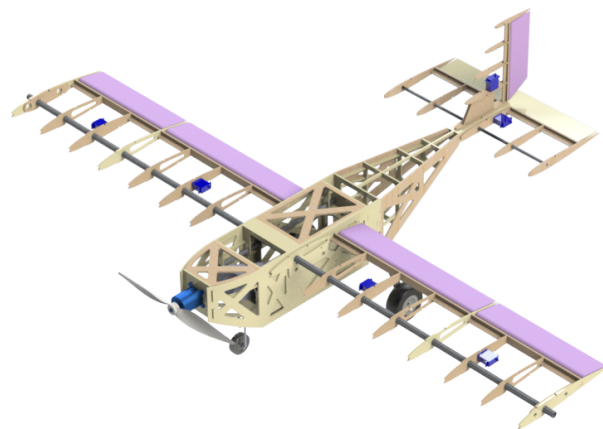


Figure 4.1: Finalized Design of *Bear Force One*

5.1 Dimensional Parameters

A CAD model of the final aircraft is shown in the figure below. Table 5.1 summarizes the key dimensions of the final aircraft design.

Table 5.1: Key Dimensions of Final Aircraft Design

Vertical Stabilizer		Horizontal Stabilizer	
Span [mm]	232.9	Span [mm]	383.5
Mean Chord [mm]	151.6	Mean Chord [mm]	201.2
Area [m ²]	0.0353	Area [m ²]	0.0768
AR	1.54	AR	1.92
Wing		Overall Aircraft	
Span [mm]	1311	Length [mm]	919.5
Mean Chord [mm]	231.6	Width [mm]	1311
Area [m ²]	0.3036	Empty Weight [kg]	1.73
AR	5.66	Max Weight [kg]	2.95

5.2 Structural Characteristics and Design Methodology

WUDBF's primary structural design goals were to maximize strength and stability, maximize accessibility to the electronics and payload compartments, minimize weight, and simplify the manufacturing process. The overall structure is designed to direct loads from each subsystem into the major load bearing components of the aircraft. The fuselage consists of wooden formers connected by longerons with two central CF rods running through each former to resist bending moments. Wooden side panels also provide torsional resistance and structural stiffness. Propulsive loads from motor thrust and torque that can cause bending, torsion, and vibrations are directed through a wooden motor mount into the fuselage. Ground loads from the landing gear are directed through the fuselage structure and into the central CF rods. The wing consists of wooden ribs connected by stringers and a spar, which provide bending and torsional stiffness. Aerodynamic loads from the wing are transferred to the fuselage side panels, where they are then distributed across the fuselage structure and the central CF rods. The empennage is attached to wooden plates that interface with fuselage longerons, distributing loads into the formers and thus the central CF rods. The figure below shows a path diagram of these various expected loads.

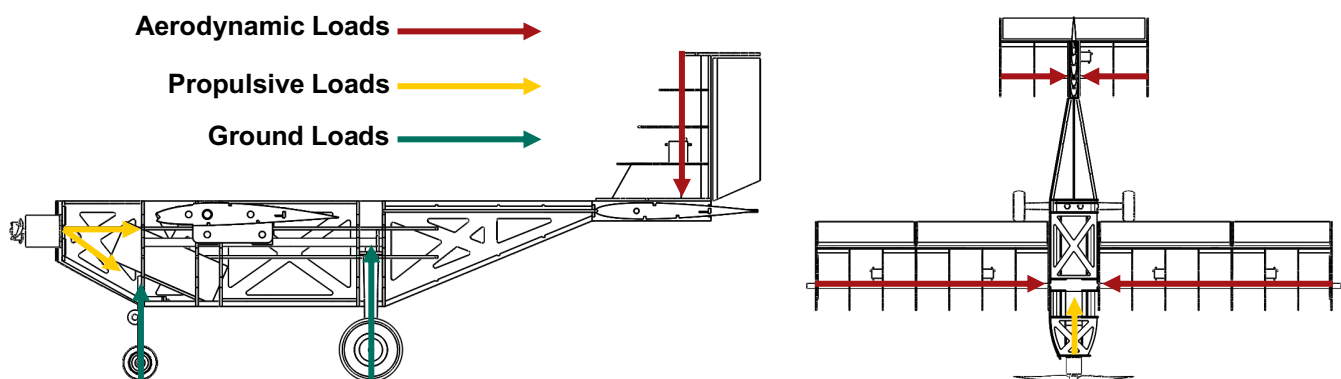


Figure 5.2: Load Path Diagram

5.3 Subsystem Design

5.3.1 Fuselage

The fuselage is made mainly of balsa wood and lite plywood. It is primarily designed to encase the electronics package, control electronics, and battery.

Also, it directly connects the motor, landing gear, wing, and tail. This design can be seen in Figure 5.3. The lite ply formers are connected by longerons stabilizing each rib. These formers connect to the lite ply sidewalls using interlocking tabs. The sidewalls start at the rear end of the payload bay and bend at the nose to connect to the motor mounting plate. One CF rod extends lengthwise through the formers on each side of the fuselage, just inside the sidewalls. A third CF rod runs crosswise and fits around the wing CF rods when the wings are attached. The back of the fuselage has easy access cuts to allow for the placement of electronics and the battery. The top plates can hinge to allow loading of the battery and electronics package. Balsa shear plates are affixed to the longerons and formers in the rear of the fuselage and distribute aerodynamic loads from the empennage to the rest of the aircraft, reducing vibration. During construction, two loops of high-strength climbing rope are looped around the crosswise spar. These can be stored in the fuselage during flight, but during the ground mission, they can extend out the bottom of the fuselage to attach to the weight holder.

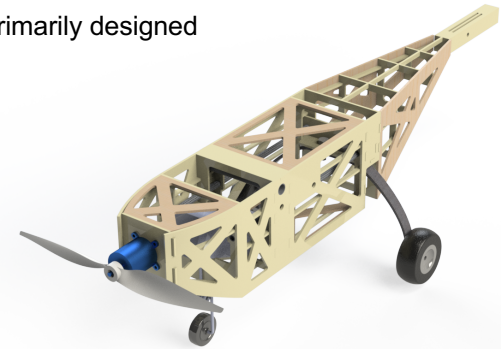


Figure 5.3: Aircraft Fuselage

5.3.2 Wing

The wing structure is made up of 16 ribs total with one CF spar running through each half-wing for structural support and to help resist twisting of the wing. The innermost rib of the half-wings have tabs to attach to the sidewalls with nylon bolts. These bolts allow for easy attachment and removal of the alternative wing sections. The wing spars extend two inches inboard of the innermost rib to provide sufficient interface with the fuselage CF rod that they slide into. Thus,

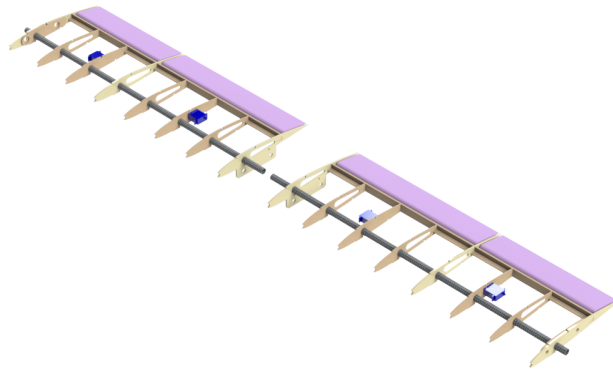


Figure 5.4: Aircraft Wing

when the wings are attached, any loads on the wings will be transferred through the spars to the fuselage CF rod, which will then distribute loads throughout the aircraft. The spars also extend one inch outboard of the wingtips to support the antenna adapters and ground test fixtures. Servo and control surface attachment points were also included in the wing structure. The control surfaces are constructed from XPS foam with a fiberglass layup. Finally, the leading edges of the ribs are wrapped with balsa wood to increase the stiffness of the wing structure. This can be seen in Figure 5.4.

5.3.3 Empennage

The empennage was designed similarly to the main wing design. The HT is supported by a single CF rod through the entire span. The VT does not have a CF rod, but two plywood tabs extend through its lower ribs and fit around the HT CF rod. Both the HT and the VT are glued to a box structure on the rear of the fuselage, which fits around the longerons to increase its stiffness. Mounting plates for control servos and the rudder and elevator hinges are also integrated into the rib structure. This structure can be seen in Figure 5.3.3.

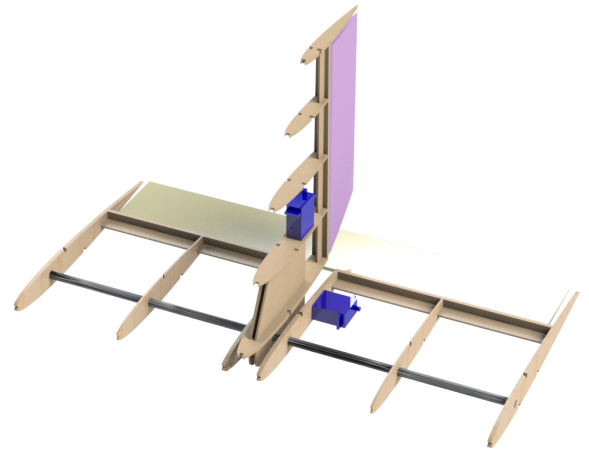


Figure 5.5: Aircraft Empennage

5.3.4 Main Landing Gear

CF landing gear was purchased online because WUDBF did not have the equipment to manufacture it. The JTC-E10 Extra Style Landing CF landing gear was chosen and placed in the rear of the fuselage. Since the landing gear was placed centrally in the fuselage instead of the bottom, a height of 5 inches was selected to give the required propeller clearance when the wheels were installed. The landing gear is bolted to a horizontal lite ply plate in the vertical center of the fuselage. This plate interfaces both with the fuselage formers, and an additional vertical lite ply plate connected directly to the lengthwise CF rods. This structure is shown in Figure 5.6. The nuts securing the gear can be removed, and the landing gear can be rotated rearward to reduce the height of the aircraft. The nosewheel is attached to a shock-absorbing metal rod, which fits into a collar screwed to one of the formers. A servo in the fuselage allows the pilot to steer the aircraft using the nosewheel.



Figure 5.6: Rear Landing Gear Attachment

5.3.5 Antenna Adapter

The antenna adapter is 3D printed to best incorporate the attachment points required in the rules. Two holes in the front and rear of the adapter all bolts to fix it to the wingtip. A third hole fits over the wing CF spar, providing extra resistance to moments generated by the antenna drag. The antenna will fit into a cutout in the top of the adapter, with a horizontal hole to accept a third bolt. This bolt will pass through a hole drilled in the bottom of the antenna, securing it to the adapter. The counterweight will have the same profile as the adapter, but will be a hollow print, allowing it to be filled with weights to balance the weight of the antenna.

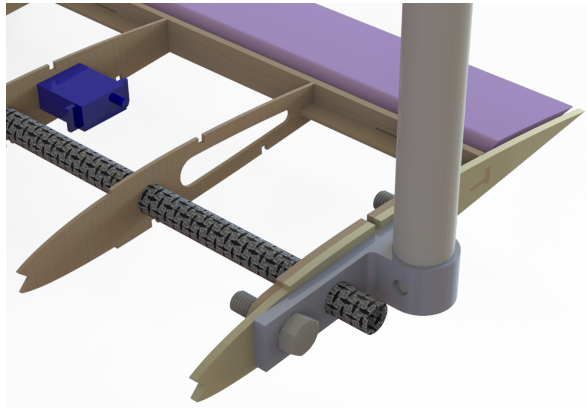


Figure 5.7: Antenna-Wing Adapter

5.3.6 Electronics Package

The electronics package is a hollow box, measuring 3x3x6". A bag filled with lead BB's is attached to the inside of the box. Since the BB's do not take up the entire volume of the package, the bag's placement inside the box can be adjusted to reduce the package's effect on the aircraft's CG. The package will have Velcro attached to the underside, which will secure it to Velcro placed in the payload bay. The fuselage top plate is hinged and secured with fiberglass tape so that it can be opened to load the package, as seen in Figure 5.8.

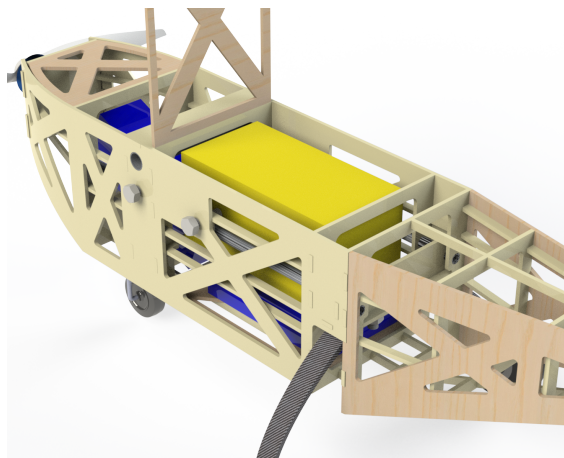


Figure 5.8: Electronics Package Storage

5.3.7 Ground Test Adapter and Stand

The ground test adapter has a similar design to the antenna adapter. It is a 3D printed device that attaches with the same two fasteners and also interfaces with the wing spar. The ground test stand consists of two T-shaped stands, each made with a wood base and aluminum supports. These two stands can be joined together by a third piece of wood to provide extra stability. The ground test adapter then fits over the top of the supports and holds the wingtip. A bucket serves as the weight holder and is attached to the rope loops with two carabiners. Weight can then be placed in this bucket.

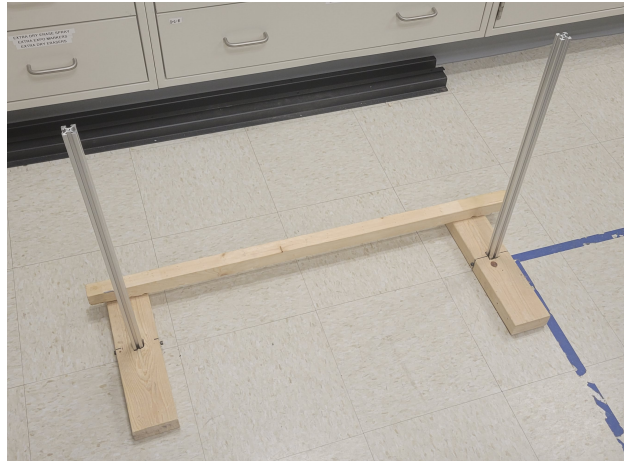


Figure 5.8: Electronics Package Storage

5.3.8 Shipping box

The shipping box consists of $\frac{1}{8}$ inch plywood sheets attached with metal L-brackets at the corners. This design allows for easy disassembly for travel, and it can be reassembled at competition. The dimensions of the box are 15.5 x 9.5 x 36.5 inches. This sums to 61.5 inches, just under the 62-inch maximum.

5.3.9 Propulsion System

The propulsion system consists of a SunnySky X3120-585 V3 motor spinning an APC 12x8 prop. Two different batteries are used for M2 and M3 in order to optimize performance. For a complete description of the design of the propulsion system, see section 4.5. Static testing in conjunction with flight testing confirmed that the propulsion system met thrust, power, and endurance requirements. The table below shows breakdowns of the propulsion system for each mission.

Table 5.2: Propulsion System Component Breakdown

Component	All Missions
Motor	SunnySky X3120-585 V3
Propeller	APC 12x8
Propulsion Battery	Admiral 4500mAh 6S 45C LiPo
ESC	Phoenix Edge Lite 75 A (8S)
Receiver Battery	Gens Ace 1300mAh 3S 45C LiPo
Receiver	FrSky RX8R 8/16Ch

5.4 Weight and Balance

Each mission has a different mass distribution. To estimate the CG of the vehicle, component masses and locations were determined from the CAD model and tabulated. The leading edge of the imaginary central wing rib was taken to be the origin, and a mutually orthogonal coordinate system was established with the positive X-axis pointing out the nose of the aircraft, positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right wing when viewing the aircraft from behind. The masses and absolute coordinates of the aircraft's major components are listed in Table 5.3. The weights and CG measurements for M2 and M3 were calculated including the electronics package and antenna for each mission respectively. Note that components such as bolts, glue, covering, and wire mass are not included in the table.

Table 5.3: Weight and Balance for Each Mission

Component	Weight [kg]	CG _x [mm]	CG _y [mm]	CG _z [mm]
Wing	0.13	-95.00	0	-3.56
Fuselage	0.29	-73.91	0	59.18
VT	0.008	-648.72	0	-77.22
HT	0.033	-657.86	5.84	-8.13
Rear Landing Gear	0.095	-269.75	0	133.10
Front Landing Gear	0.04	17.53	0	129.29
7x Control Servos	0.089	-242.57	5.08	-1.27
ESC	0.085	73.15	0	-73.15
Control Electronics	0.15	87.12	0	-87.12
Propulsion Battery	0.65	-21.08	0	55.63
Motor	0.16	133.35	0	16.76
M1				
Overall	1.78	-104.65	6.35	50.55
M2				
Electronics Package	1.22	-168.66	0	26.16
Overall	3.00	-130.3	0	40.64
M3				
Antenna	0.206	-68.68	665.73	-415.04
Counterweight	0.206	-68.68	-665.73	-3.56
Overall	2.19	-46.99	6.35	-106.17

5.5 Structural Performance

In order to verify structural performance of the aircraft, WUDBF utilized Solidworks finite element analysis (FEA) to simulate the major structural components of the aircraft under their maximum expected load conditions.

5.5.1 Wing Spar FEA

FEA was performed in Solidworks on the CF spar in the wing to ensure it could withstand a wingtip loading test with an approximate load factor of 2.5g. The finite lift distribution provided by XFLR5 was assumed to be near elliptical. In order to input a continuous lift distribution into Solidworks, curve fitting was performed in MATLAB as shown in the figure below.

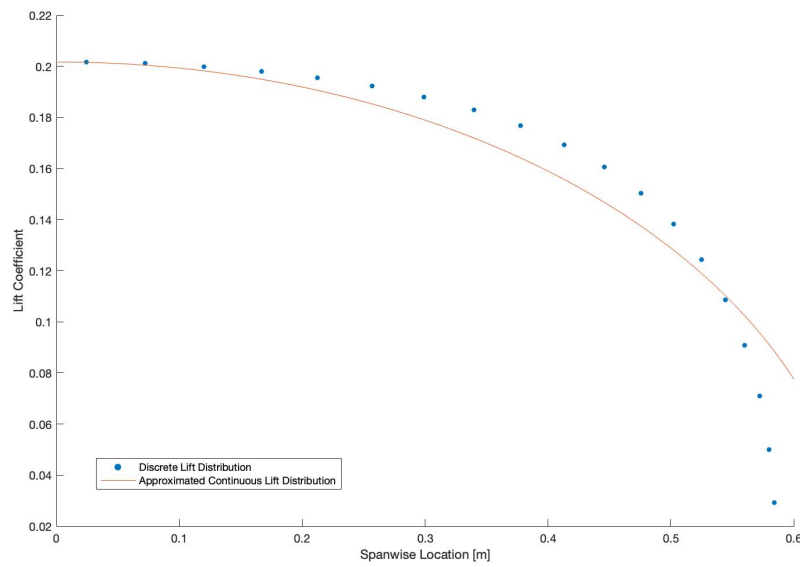


Figure 5.10: Finite and Continuous Spanwise Lift Distribution

Using the continuous lift distribution and applying a load factor of 2.5g, the following von Mises stress plot was generated using Solidworks.

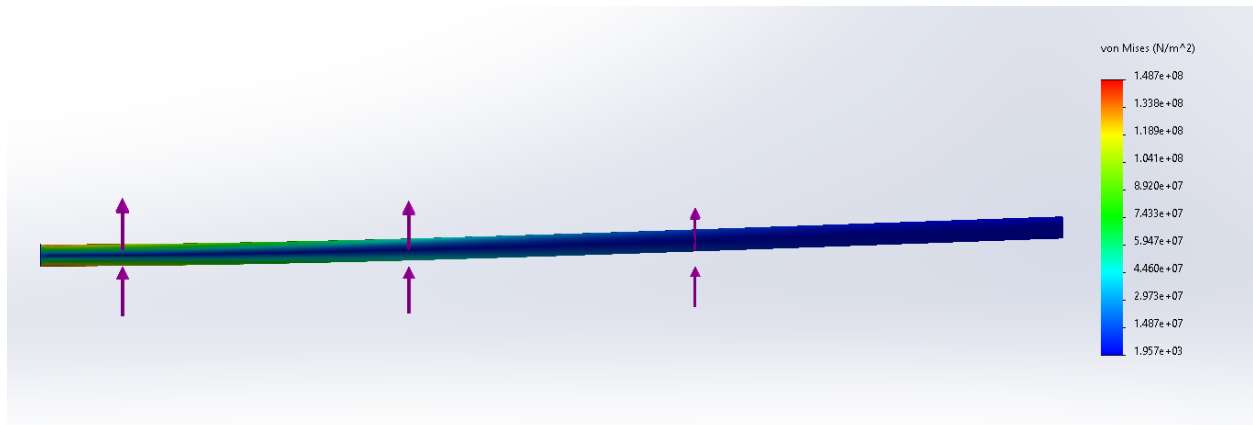


Figure 5.11: Von Mises Stress Plot of the Wing CF Spar Experiencing a Load Factor of 2.5

Based on the simulation, the maximum von Mises stress experienced by the CF spar is 149 MPa. Given a tensile strength of 3450 MPa [10] for standard modulus carbon fiber, this results in an adequate factor of safety of 23.2. Based on the FEA, the 2.5g load case also produced a wingtip displacement of 15.7 mm which was considered reasonable given that the actual displacement of the spar would be lower with the addition of the wooden structure around it.

5.5.2 Main Landing Gear FEA

FEA was also performed on the fixed, CF rear landing gear to ensure it has the structural integrity to survive landing. The fully loaded M2 weight was applied to the top surface of the gear with a loading factor of 3g to model a typical landing. A von Mises stress plot was generated as shown in the figure below.

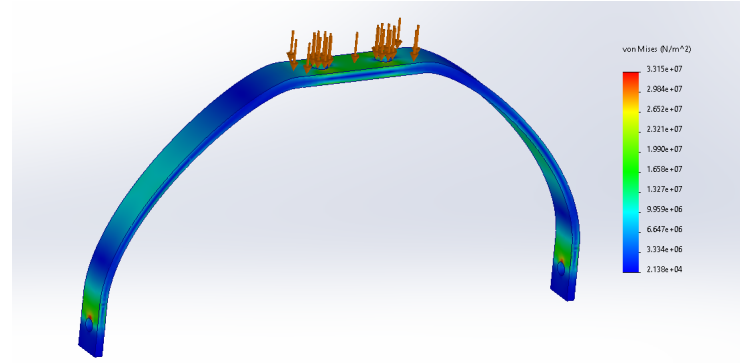


Figure 5.12: Von Mises Stress Plot of the Main Landing Gear Experiencing a Load Factor of 3

As shown in Figure 5.12 the max von Mises stress was 22.1 MPa. Given a maximum tensile strength of 1896 MPa for carbon fiber, this gives a more than adequate factor of safety of 85.8 [11]. Additionally, the maximum displacement experienced by the landing gear was 0.005 in, which was acceptable since there was 6.5 in of designed prop clearance.

5.6 Predicted Final Aircraft Performance

Using the updated weights in Table 5.4, Km_sim was used to simulate the expected flight and mission performance of the final aircraft. The extracted performance estimates are shown in the table below. To obtain normalized scores, best scores for M2 and M3 were estimated based on Section 3.3.

Table 5.3: Predicted Final Aircraft Performance

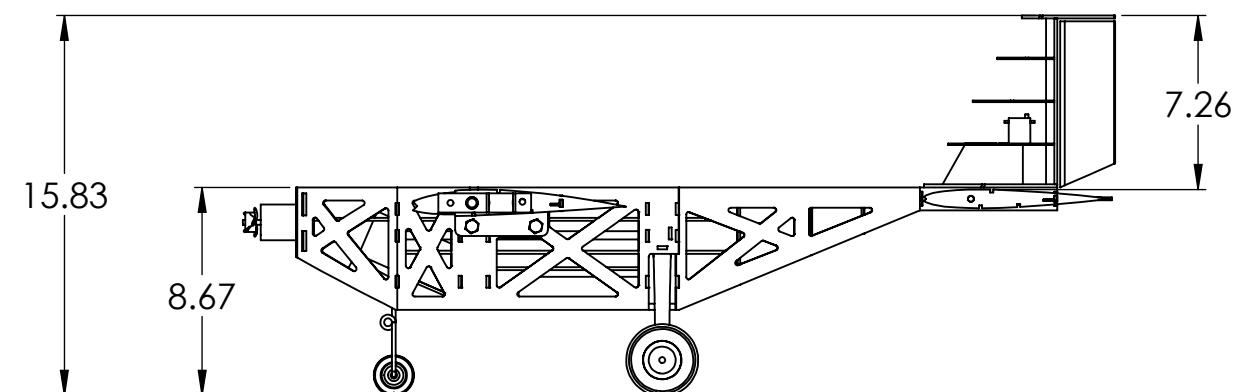
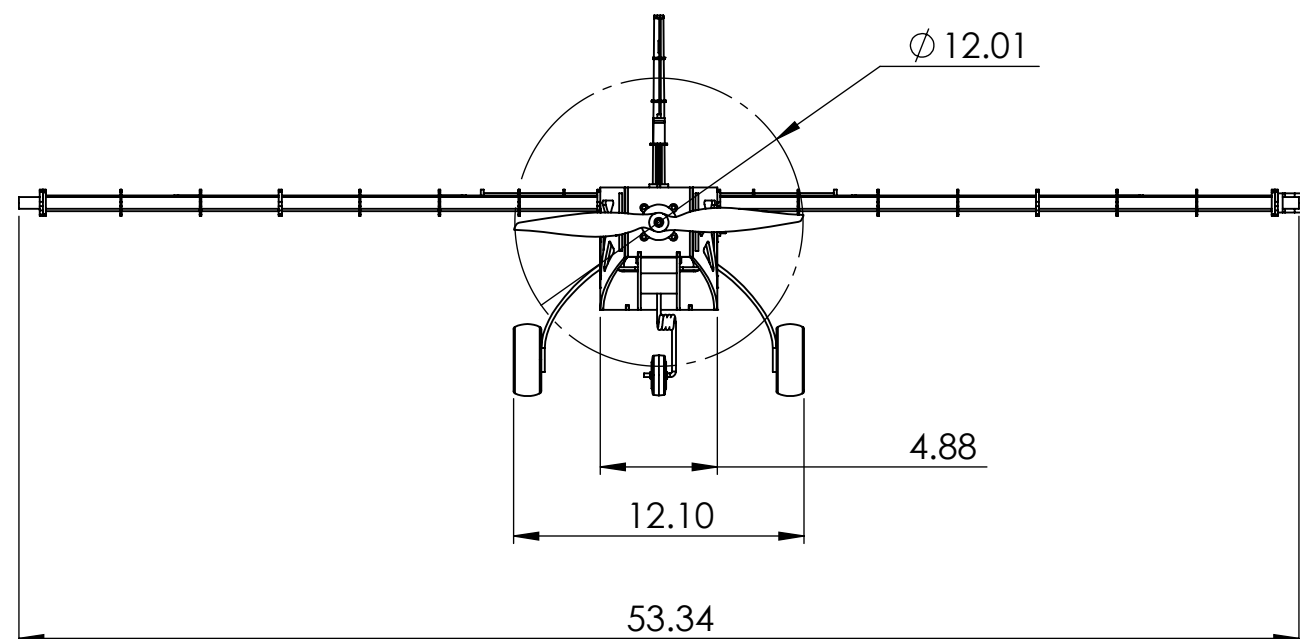
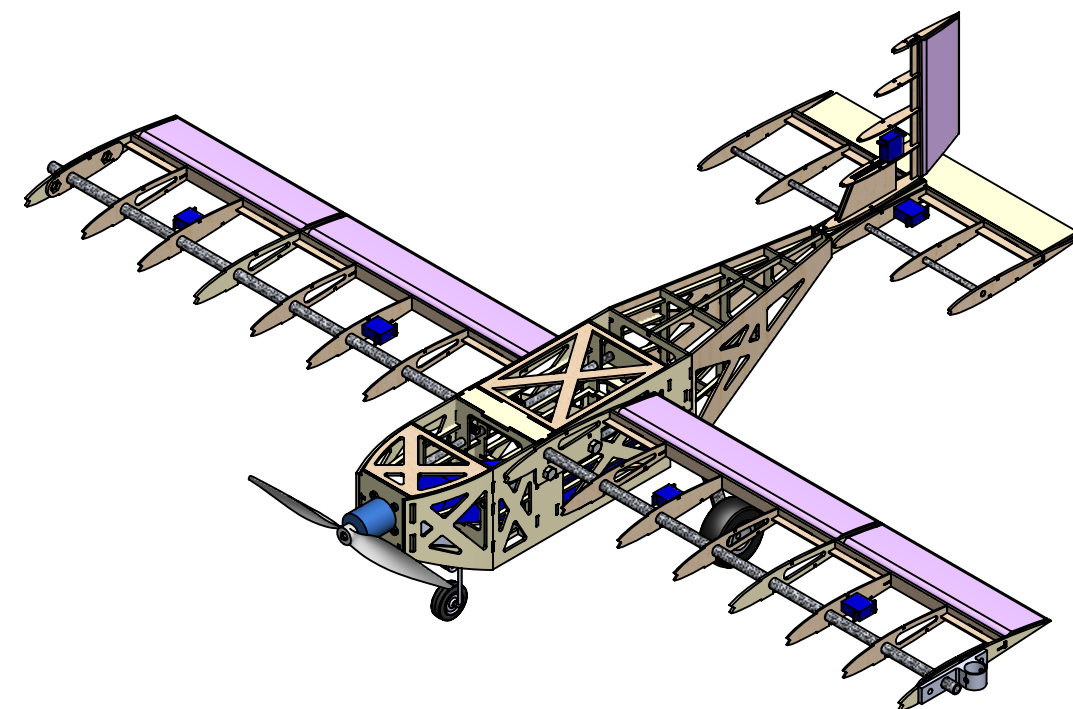
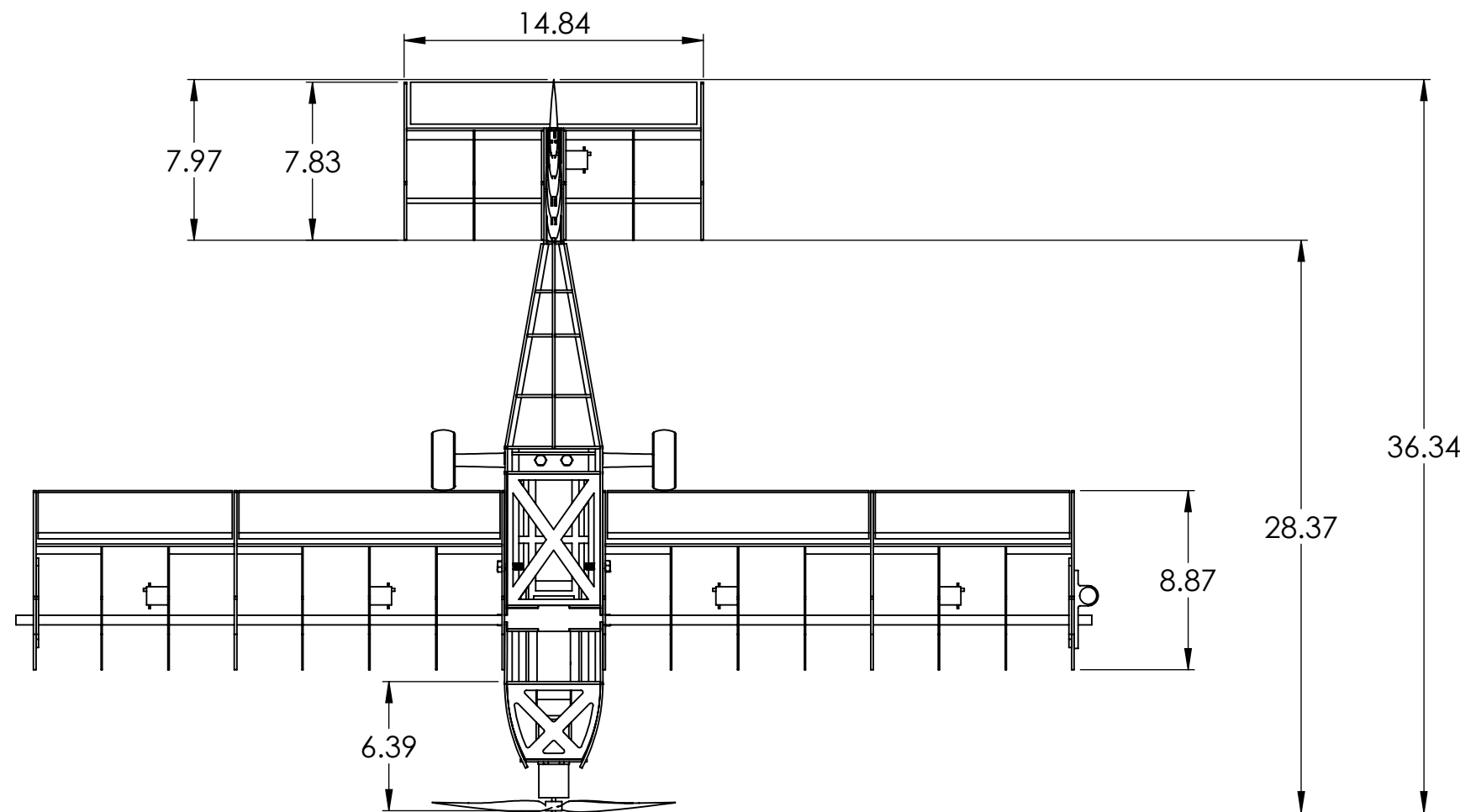
	Mission 1	Mission 2	Mission 3
Inputs			
All-Up Weight [kg]	1.78	3.01	2.19
Payload Weight [kg]	-	1.23	0.41
Wing Loading [kg/m ²]	6.51	11.01	8.01
Antenna Length [m]	-	-	0.86
C _{Lmax}	1.735	1.735	1.735
Parasitic Drag	0.05611	0.05611	0.13676
Cruise Throttle	1	1	1
Turn Load Factor	2.5	2.5	2.5
Outputs			
Takeoff Distance [ft]	18.08	32.57	23.27
C _{Lcruise}	0.18338	0.37875	0.18338
C _{Dcruise}	0.00214	0.00905	0.0024
Cruise Velocity [m/s]	37.78	37.70	30.76
Number of Laps	3	17	3
Mission Time at End of Final Lap [min]	1.75	9.83	2.19
Scoring			
Bear Force One Score	1	205.06	39.27
Normalized Score	1	1.46	2.42

4

3

2

1



All dimensions are in millimeters

Washington University in St. Louis
AIAA Design Build Fly 2023

Name:

Bear Force One

Aircraft 3-View

SCALE: 1:8

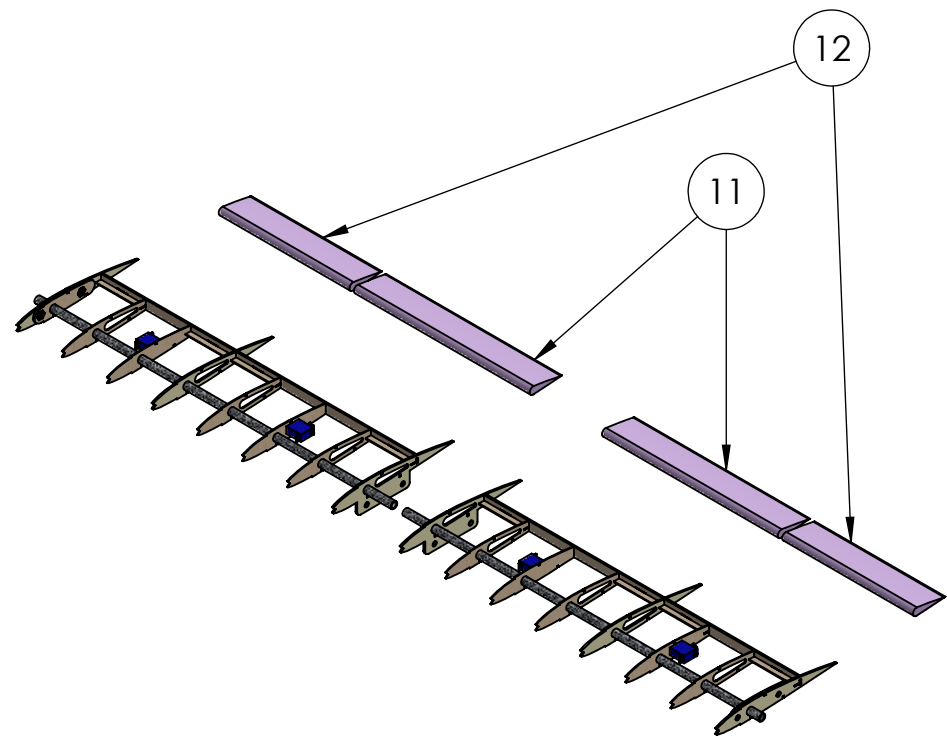
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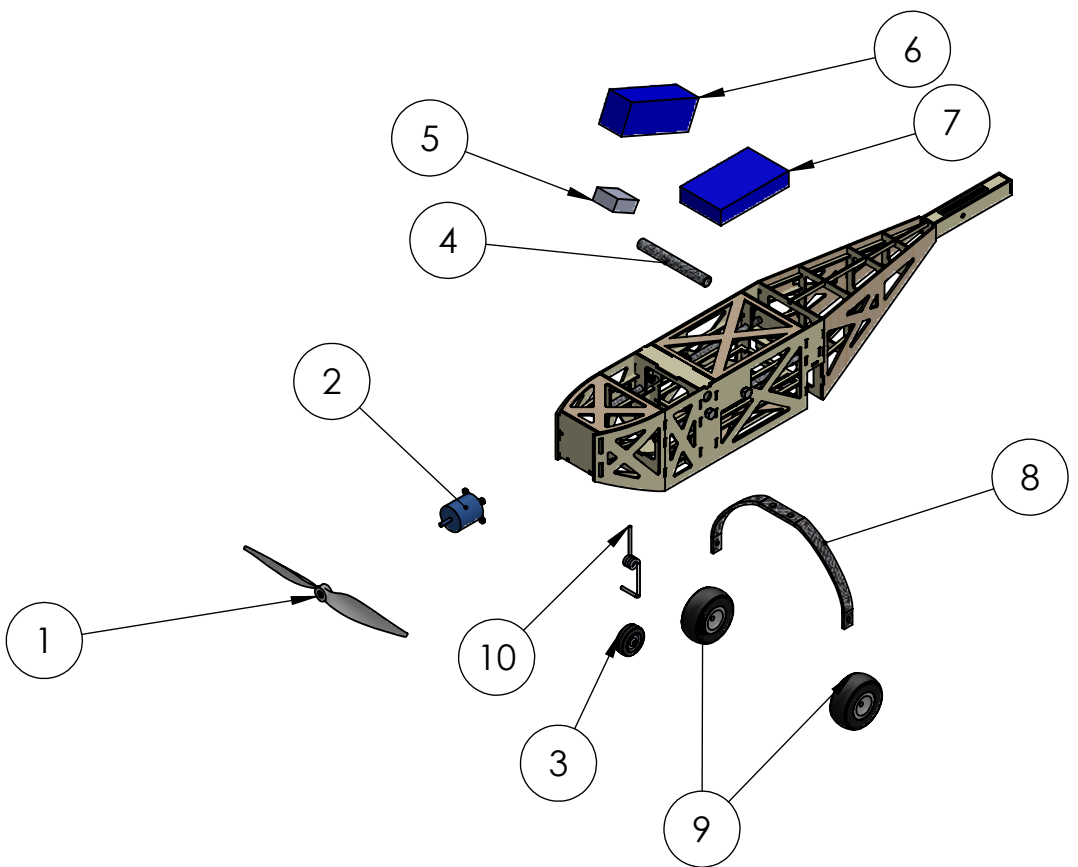
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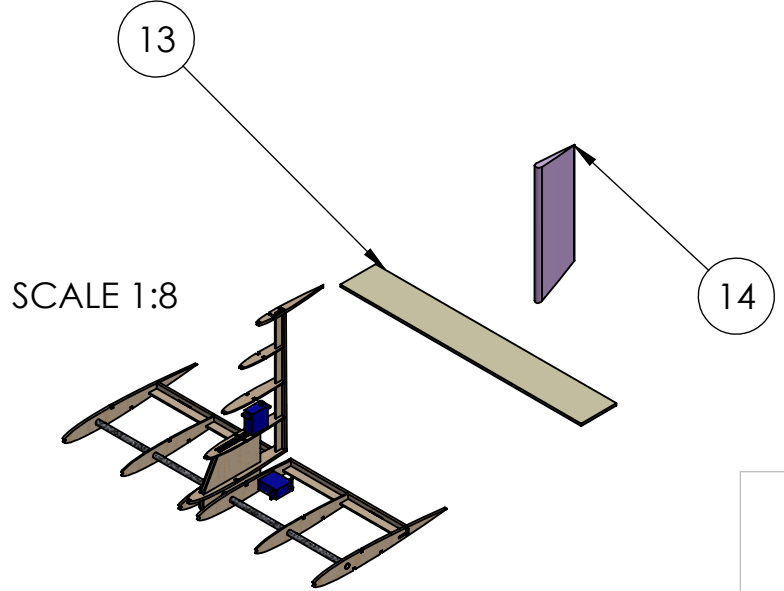
Bill of Materials

Item No.	Item Name	Qty.	Material
1	Propeller	1	Plastic
2	Motor	1	Aluminum
3	Front Landing Gear Wheel	1	Foam
4	Wing CF Connection	1	Carbon Fiber
5	ESC	1	-
6	Battery	1	Lithium Polymer
7	Control Electronics	1	-
8	Rear Landing Gear	1	Carbon Fiber
9	Rear Landing Gear Wheels	2	Nylon
10	Front Landing Gear Spring	1	Aluminum
11	Flaps	2	XPS Foam
12	Ailerons	2	XPS Foam
13	Elevator	1	Balsa
14	Rudder	1	XPS Foam

A



B



SCALE 1:8

A

Washington University in St. Louis
AIAA Design Build Fly 2023

Name:
Bear Force One

Aircraft Exploded Layout

All dimensions are in millimeters

SCALE: 1:10

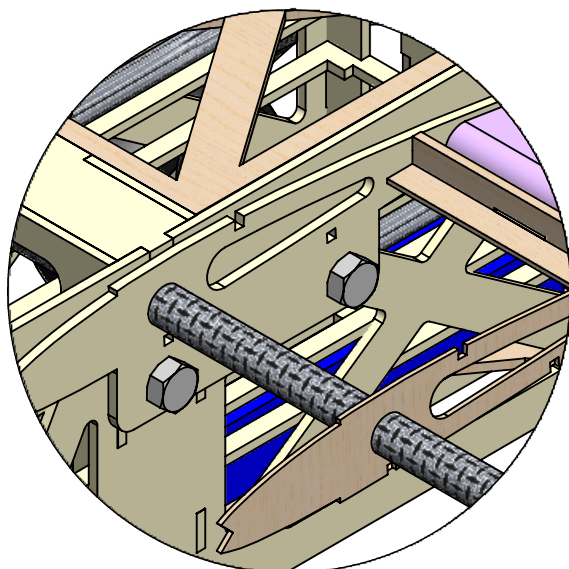
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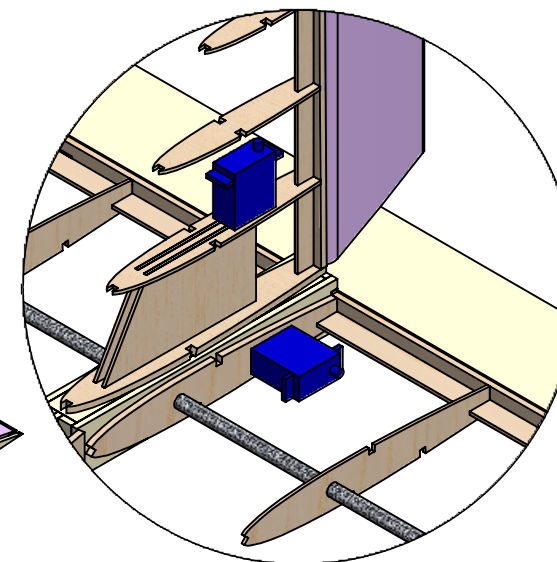
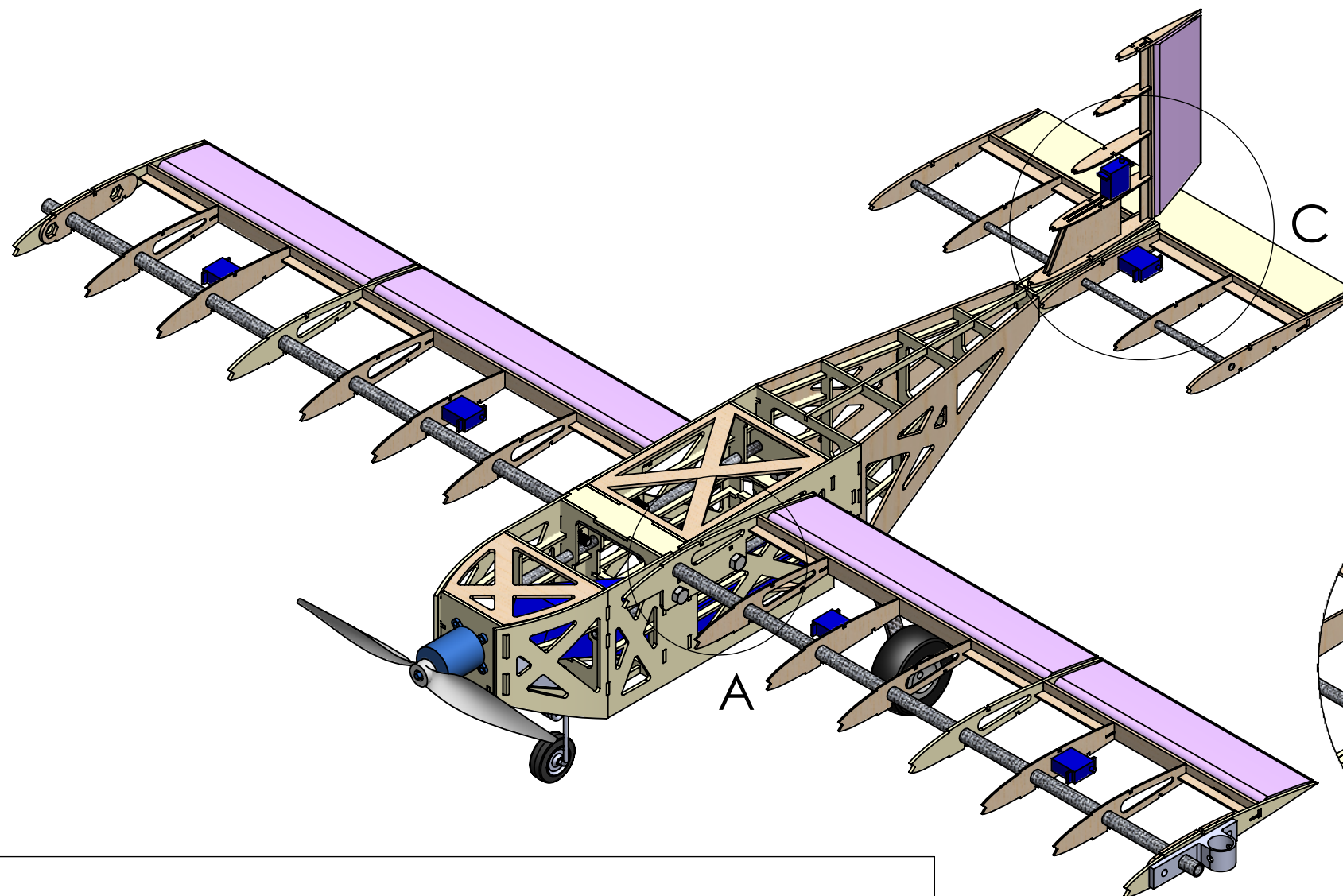
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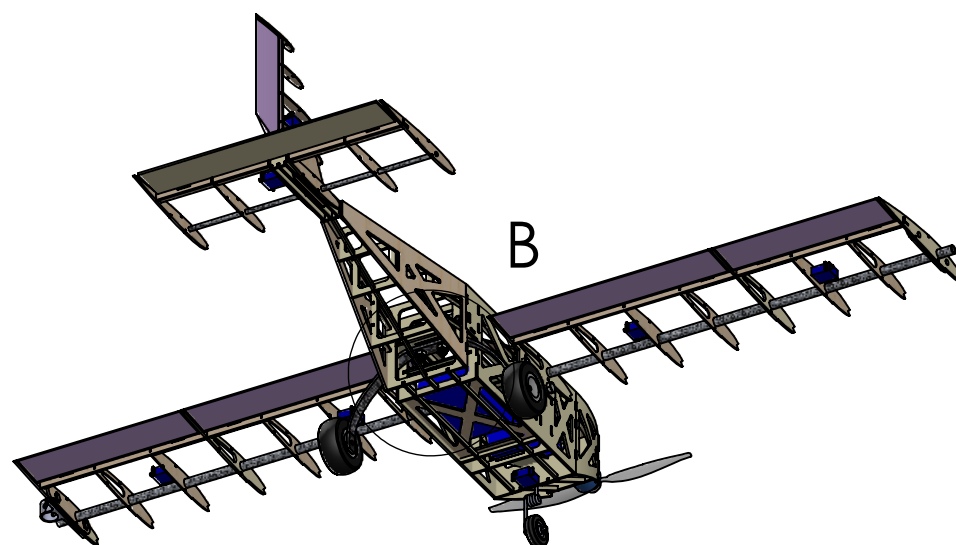
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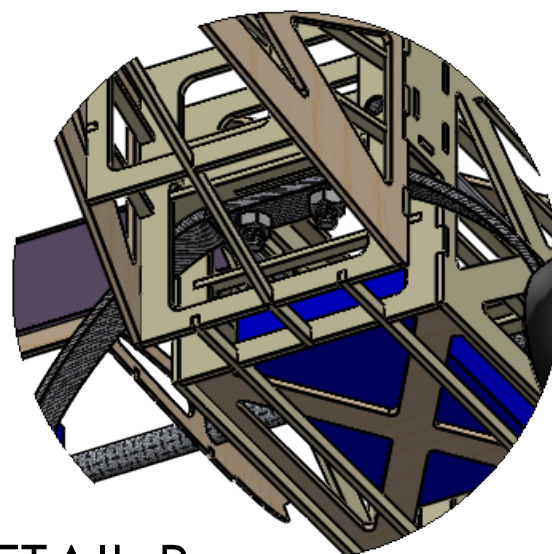


DETAIL C

SCALE 1 : 3



SCALE 1:10



DETAIL B

SCALE 1 : 3

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AIAA Design Build Fly 2023

Name:

Bear Force One

Aircraft Detail

SCALE: 1:5

SHEET 3 OF 4

3

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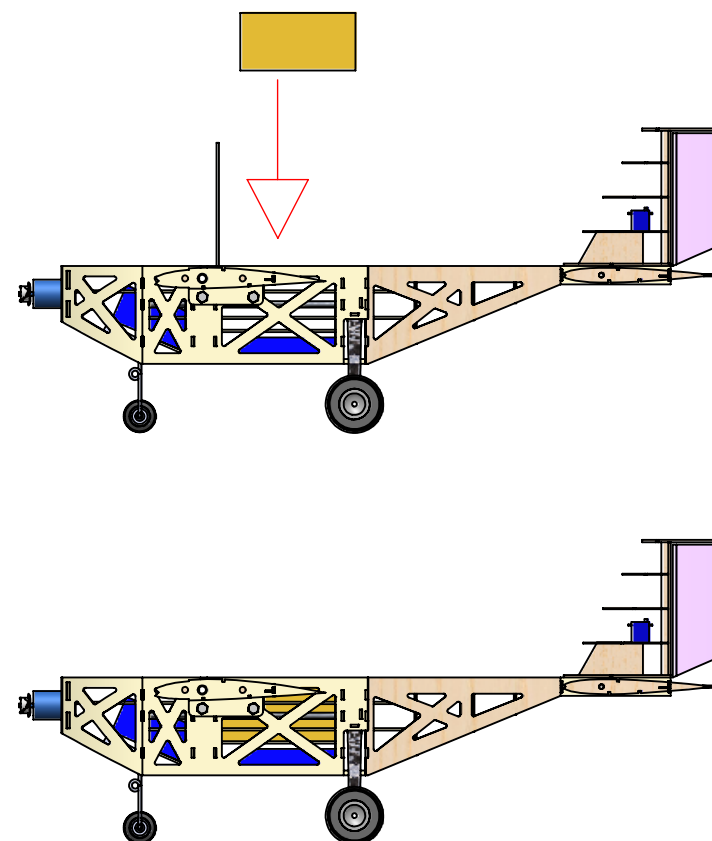
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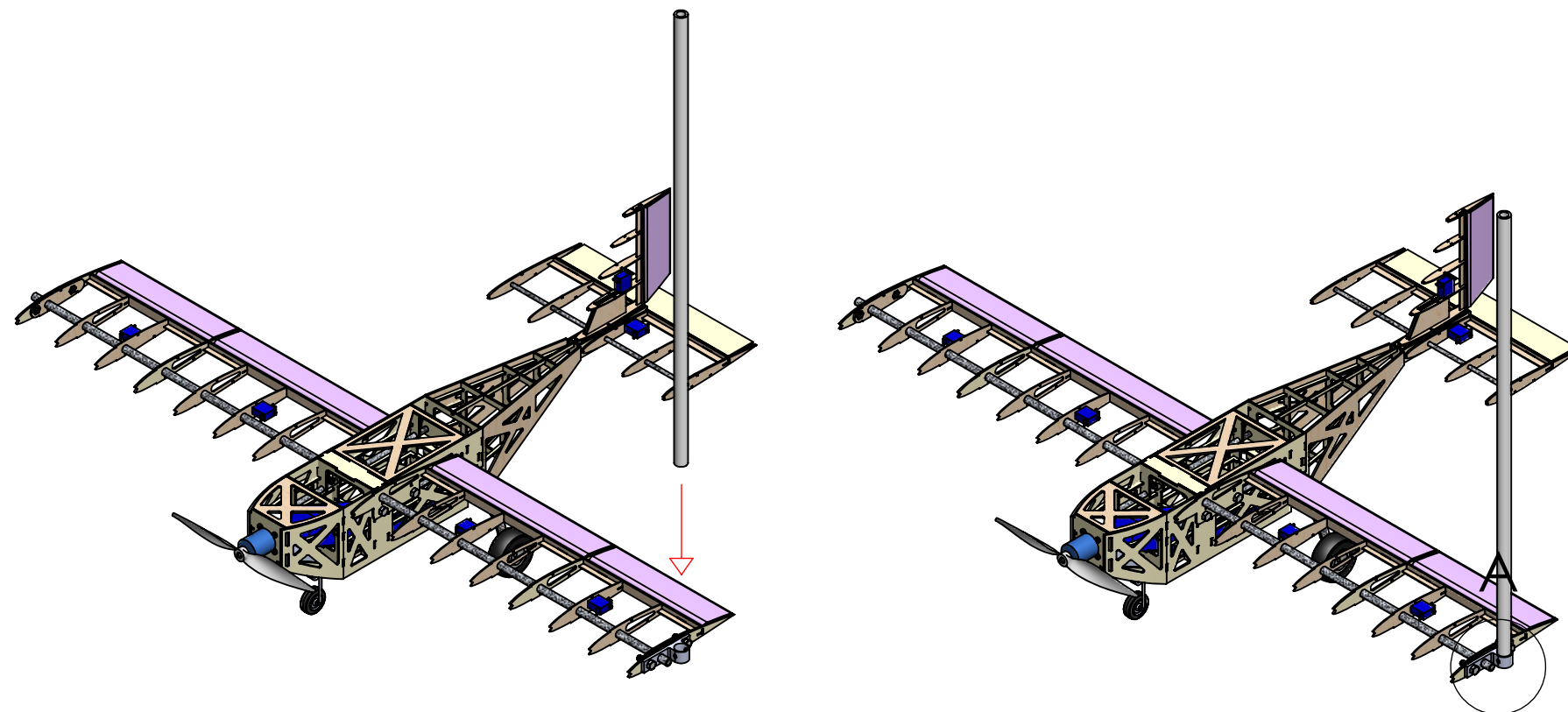
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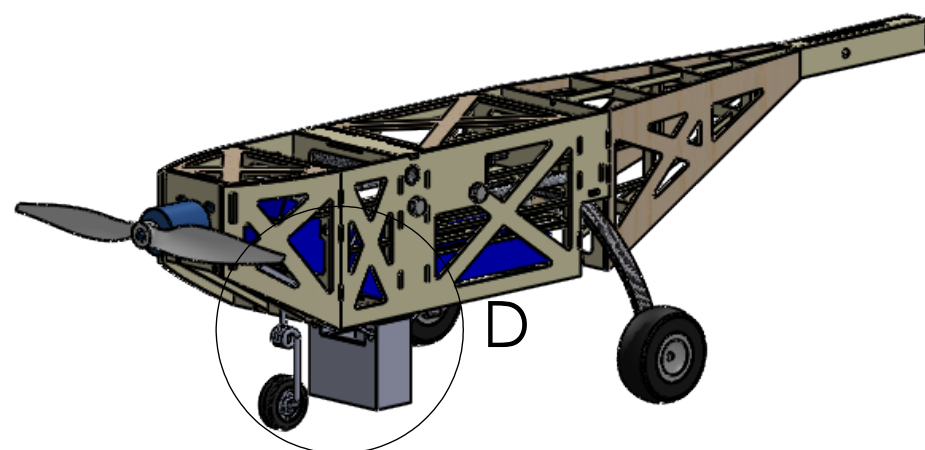
Electronic Package Insertion
(ARROW DESCRIBES MOTION OF THE PACKAGE)



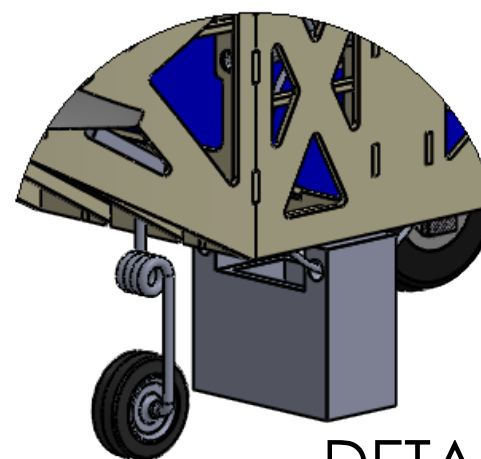
Antenna Insertion
(ARROW DESCRIBES MOTION OF THE Antena)



Ground Mission Test Weight Bucket
(INSERTED FROM BELOW)



SCALE 1:6

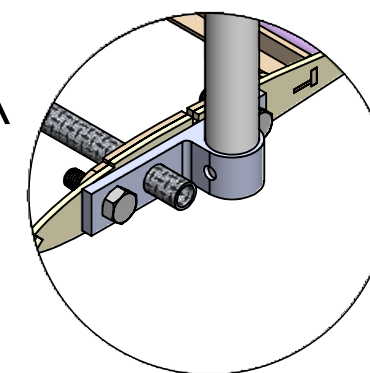


DETAIL D

SCALE 1 : 3

DETAIL A

SCALE 1 : 3



Washington University in St. Louis
AIAA Design Build Fly 2023

Name:

Bear Force One

Mission Accomodations

SCALE: 1:10

SHEET 4 OF 4

3

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6. Manufacturing Plan

6.1 Selection of Manufacturing Processes

Multiple manufacturing processes were considered to identify the best method for constructing the aircraft and the various subsystems.

6.1.1 Selection Criteria

Key evaluation criteria included ease of construction, effectiveness, feasibility, and weight. Materials which had been used in past aircrafts were generally preferred, given manufacturing methods for those materials had already been refined. Cost was also considered due to the club's limited budget.

6.1.2 Solid Foam

Solid foam components are able to be designed and fabricated quickly since there are no internal bracing structures. Using foam along with carbon fiber and fiberglass sufficiently strengthens the foam. Additionally, the use of a hot-wire cutter to accurately shape the material makes it a desirable material to form smaller, precise parts of the aircraft such as the ailerons, rudder, and elevator. Other advantages of using foam to fabricate parts of the aircraft include its light weight.

6.1.3 Additive Manufacturing

3D printing has advantages during construction due to the ease of fabricating structures as single components and the streamlined workflow from CAD to printer. However, 3D printers are slower than other methods for constructing large shapes. 3D printed parts are also heavier and weaker than those made from other materials, especially in shear along the seams between layers. While WUDBF ultimately did not make use of 3D printed parts for the main aircraft structure, more intricate components that don't bear significant loads such as servo mounts were 3D printed due to the flexibility and ease of 3D printing.

6.1.4 Composite Layup

Fiberglass or CF composite layup methods were considered due to their high strength and low weight. These methods however require construction of molds or plugs, which were determined to be too time consuming and difficult to manufacture with current resources. WUDBF also has limited experience with these methods. That being said, prefabricated CF rods and fiberglass layups were used to reinforce the major load bearing components on the aircraft.

6.1.5 Wood Construction

Wood construction is the manufacturing technique WUDBF is most experienced with. WUDBF has a laser cutter that can be used to accurately and rapidly cut out wooden parts. A number of different woods are used, primarily balsa, basswood, and birch ply. Each type of wood offers different material properties. This creates flexibility when choosing where weight can be saved or where extra strength is needed. Assembly of wooden components can be done using CA glue and epoxy. CA glue dries quickly and forms a strong bond, making it the primary adhesive in wooden construction. Two-part epoxy offers a stronger alternative for components that bear more structural loads. In order to maintain the aerodynamic shape of the aircraft, the frame can be covered in Monokote, a heat-shrinking plastic film that can be tacked onto

the aircraft and then heated, pulling it tight across the frame. Monokote is relatively thin and susceptible to tearing but can be quickly and easily repaired.

6.1.6 Processes Selected for Major Components

Wood construction, with Monokote coating, was chosen as the primary manufacturing method for all major components due to its structural qualities and WUDBF's previous experience. All wooden components including wing ribs and fuselage formers are cut using a laser cutter. These pieces are then assembled using CA and epoxy. CF parts and fiberglass reinforcement are added in key areas for structural support. In particular, the central rod, wing spars, and landing gear are CF parts purchased online, and the wood plates connecting the landing gear to the fuselage are reinforced with fiberglass.

6.2 Detailed Manufacturing Processes

The manufacturing processes used in the construction of the aircraft are detailed below. Note that these descriptions are primarily based on the construction of the prototype aircraft and some adjustments may occur for the final aircraft, which is currently being built.

6.2.1 Wing and Tail

The wing is composed of ribs, strings, and a main CF spar. Each half-wing is constructed separately and is designed to be easily removable. Each CF spar slides into a secondary CF rod located in the fuselage. The ribs are aligned using slotted basswood jigs, with each rib fitting into a designated slot. Once the ribs are aligned, the CF spar is inserted and secured using epoxy/CA. A quarter-inch square basswood stringer is then sanded and attached to the leading edge to achieve the correct curve. Next, the mounting plates for the control surfaces are installed, the jigs cut off and sanded down, and the wing flipped over. Basswood stringers are added. The leading edge is then formed by wrapping sheets of balsa soaked in warm water around the leading edge stringer. 3D-printed servo mounts are then glued to the ribs for later electronics installation. Control surfaces are constructed from foam and reinforced with fiberglass. Hinges are then inserted and epoxied into the control surfaces and onto the hinge plates. Finally, the wing and control surfaces are covered with Monokote.

The tail is constructed using basswood stringers, balsa wood, and thin CF rods. Like in the wing, basswood stringers serve as the leading edges. These stringers are sanded to the shape of the desired airfoils. Each component of the tail is constructed separately in an identical manner to the wings, using similarly designed jigs. The elevator and rudder are also constructed separately and then attached to the HT and VT respectively, using hinges and CA glue. To connect the tail to the fuselage, holes oriented perpendicularly to each other are drilled into the central CF rod, which extends out of the fuselage and acts as the tail boom. A custom hole drilling jig is used to ensure that the holes are perpendicular to each other and aligned properly with the wings and the rest of the aircraft. Next, a rectangular box made of laser-cut birch ply is assembled and glued in the proper position at the end of the tail boom. A thinner CF

rod connected to the tail is slid through the box and CF rod. Then, the VT and HT are epoxied to the box. After the installation of servos and servo mounts, the tail is covered in Monokote.

6.2.2 Fuselage

The fuselage is constructed primarily from birch ply formers, four panels also made from birch ply, and basswood stringers. It was found that balsa stringers warp under the force applied by Monokote, so basswood stringers are used to prevent any deformation. Each side panel has puzzle piece-like connectors for ease of construction. The panels also have slots for the formers to slide into. With everything snapped into place, the stringers and interlocking sides are CA glued into place. The central CF rods are then inserted and epoxied to the formers. Two basswood landing gear plates are then installed. These plates are reinforced with fiberglass for extra strength. The CF landing gear is sandwiched by these plates and secured using nylon bolts. The nose mount contains the motor and is constructed separately from the main fuselage. The nose snaps into the main fuselage body using the same jigsaw connectors. An additional metal plate for the motor mount is installed for reinforcement. Once assembled, everything is secured using CA glue. Following the construction of the frame and installation of electronics, the entire structure is covered in Monokote.

6.2.3 Landing Gear

The landing gear is made of CF and prefabricated. Next, the landing gear is secured using nylon bolts and basswood plates reinforced with fiberglass to withstand impact. A tricycle setup was chosen, with the front wheel allowing a small amount of freedom for steering.

6.2.4 Electronics Packae

The electronics package is constructed as a box filled with lead weights. The walls of the box are made of basswood. The box is secured with CA glue and the lid is removable so that the weight is changeable. Zinc-plated, ultra-smooth BBs are used as weights in the box. This allows the CG to remain adjustable.

6.2.5 Test Stands

The test stand is manufactured from 2x4 planks and T-slot aluminum frames. A square hole is drilled out and sanded to shape. Next, the t-slot is inserted into the hole and bolted in. Additionally, the holes are filled with epoxy for extra reinforcement. This is repeated twice to construct two sides for the test stand. Finally, a cross-beam is installed for stability. The wingtip fixtures attach to each side of the wing and slide on top of each aluminum frame.

6.2.6 Wingtip Fixtures

Two different styles of wingtip fixtures were made, one for GM, and one for M3. Each fixture uses two bolts as fasteners. The bolts are attached to the side of the wing. The M3 adapter features a cylindrical shell for the PVC pipe to slide in. There is an additional bolt hole to fasten the PVC pipe to the wingtip fixture. For the GM adapter, the t-slot frames slide into a square cap that the aircraft will rest on. Both wingtip fixtures are constructed using 3-D printing as it allows lots of iterations in a short amount of time while still remaining a strong material. The M3 adapter will use resin 3-D printing for additional strength and accuracy.

6.3 Manufacturing Milestones

A manufacturing schedule was implemented to ensure that components were constructed on time. The construction of the prototype and final aircraft were divided into important subtasks which were then used to create the schedule shown in Figure 6.2 below.

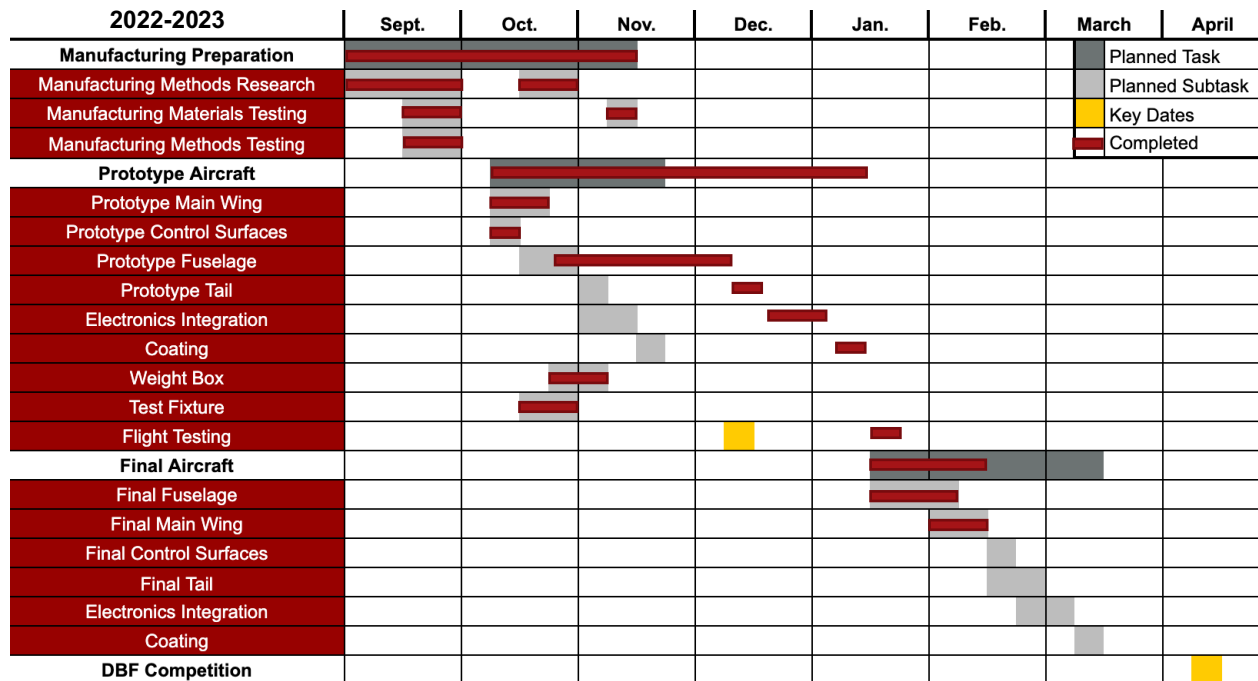


Figure 6.1: Gantt Chart of Manufacturing Schedule for The Prototype and Final Aircraft

7. Testing Plan

Ground and flight testing of the aircraft were performed to validate the design and drive iterations toward the final aircraft design.

7.1 Overall Test Objectives

Primary test objectives are listed in the table below by category. These objectives were achieved through a number of ground and flight tests described in Sections 7.2 and 7.3 respectively.

Table 7.1: Breakdown of Test Objectives

Category	Objective
Propulsion	Collect data from static thrust tests to confirm propulsion performance on the ground. Validate the propulsion system in the air during flight tests to confirm 60 ft takeoff capability, cruise velocity, and endurance
Manufacturing	Investigate manufacturing processes, in particular the benefits of fiberglassing wood components.
Structures	Conduct wingtip loading tests at maximum AUW to verify structural integrity. Perform torsion and flutter tests to ensure that deflections are not excessive. Confirm integrity of the landing gear via ground tests.
Takeoff	Confirm aircraft can takeoff within 60 ft.
Deployment	Record flight data for analysis to confirm performance requirements are met. Obtain feedback from pilot regarding handling qualities.
Performance	Confirm predictions about the Jamming Antenna drag.

7.2 Ground Testing

7.2.1 Propulsion Test

Static thrust testing was performed on the chosen propulsion system using an RCbenchmark Series 1585 Dynamometer mounted to a thrust stand. Using RCbenchmark software, thrust, RPM, current draw, and electrical power measurements were collected at incremental throttle levels. The data were then used to validate predicted performance and confirm the propulsion system met the necessary requirements.

7.2.2 Bending Test

The design requirements forced WUDBF to focus on making the aircraft as light as possible. Material choices in previous years had often been based on rule-of-thumb and precedent, so WUDBF decided to investigate the different strengths and weights of various woods to best optimize the aircraft structure. Rectangular pieces of balsa, basswood, birch ply, and lite ply were prepared. Half of the pieces were fiberglassed using 6 oz fiberglass. The test specimens were then set up as simply supported beams with force applied to the center of the span, simulating loads expected from landing or during the ground test mission. Each specimen was tested until failure, with data recorded with a load cell.

7.2.3 Wingtip Test

A wingtip loading test was performed at maximum takeoff weight to validate the structural integrity of the fuselage. This test simulated a 2.5g load case and also verified that the aircraft could survive the wingtip test required at competition. Results from the wingtip test were compared against the FEA performed in Section 5.5.2 to further validate the design.

7.2.4 Ground Mission Test

A mockup was created of the wing spars and wing attachment points in the fuselage. This was then attached to the ground test fixture and loaded to what WUDBF anticipated the final test weight to be. The structural integrity of the mockup as well as the method of applying weights was verified.

7.2.5 Antenna Test

To confirm the antenna drag prediction performed in Section 4.4, WUDBF designed and built a test fixture to measure the force on the antenna with a load cell. The antenna was attached to then attached to this

fixture, and the fixture was mounted to the side of a car. Driving at night to ensure the roads were empty, WUDBF measured the drag force on the antenna at various speeds. This data was then plotted and compared to predicted values.

7.2.6 Adapter Test

Once the force on the antenna was known, the antenna was mounted in the adapter and the drag force was simulated using a force gauge. WUDBF ensured the adapter would not break under the anticipated drag loads.

7.2.7 Box Test

A foam mockup of the *Bear Force One* was constructed to verify that the disassembled aircraft could fit inside the required box dimensions.

7.3 Flight Testing

A test flight was conducted to verify that the aircraft met design goals. A Pixhawk flight computer was used to capture telemetry during each flight test (see figure below). This data was then analyzed to extract measures of aircraft performance including speed, endurance, and lap times. Feedback from the pilot was also used to evaluate the stability and flight characteristics of the aircraft. Conclusions from flight testing were used to inform changes to the design going from the prototype to the final aircraft. For the full details of the flight test, see the test schedule in Section 7.4.

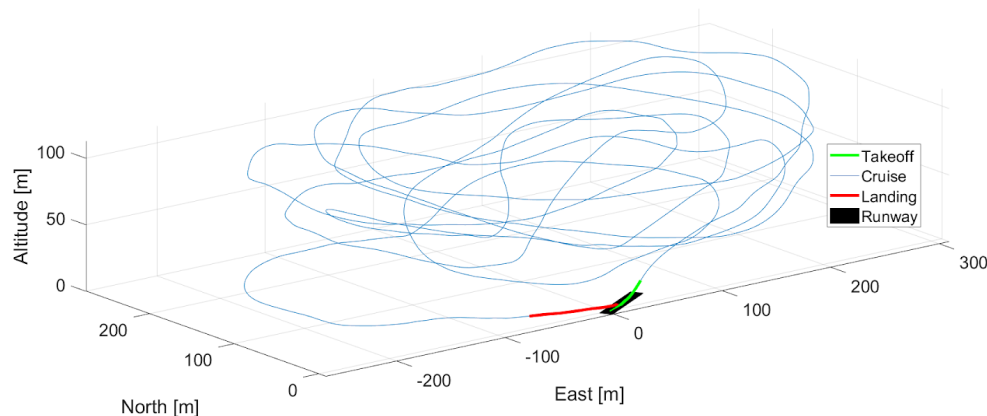


Figure 7.1: Sample Telemetry Data from Test Flight

7.3.1 Ground Inspection and Pre-Flight Checklists

Table 7.2: Ground Inspection Checklist

Component	Tasks
Motor	☐ Motor secured ☐ Propeller free from damage
Fuselage	☐ No cracks or tears ☐ Electronics are connected and secured
Wing	☐ No cracks or tears ☐ Bolted to fuselage ☐ Wingtip test
Payload	☐ Payload is positioned correctly and secure
Control Surfaces	☐ No cracks or tears ☐ Servos connected to receiver ☐ Control surfaces can deflect fully
Empennage	☐ No cracks or tears ☐ Control surfaces move without interference
Landing Gear	☐ Front wheels secured ☐ Rear wheel alignment check

Table 7.3: Pre- and Post-Flight Checklists

Flight Checklists
Pre-Flight
☐ Payload secured ☐ Propulsion battery level check ☐ Receiver battery level check ☐ CG position confirmed ☐ Wingtip test ☐ Receiver turned on ☐ Control surfaces test ☐ Landing gear steering test ☐ Range Check ☐ Motor run-up ☐ Fuselage lid is attached and secure
Post-Flight
☐ Throttle down ☐ Arming plug removed ☐ Propulsion battery disconnected ☐ Receiver battery disconnected

7.4 Testing Schedule

Table 7.4: Testing Schedule

Date	Type	Category	Objective
Prototype			
10-10-2022	Ground	Manufacturing	Test fibreglassing of wooden components.
11-13-2022	Ground	Propulsion	Static thrust test to confirm propulsion performance.
11-20-2022	Ground	Manufacturing	Verify dimensions fit within the shipping box.
1-12-2022	Ground	Structures	Wingtip test to confirm structure.
1-28-2022	Flight	Performance	Shakedown flight. Test Prototype design. (CRASHED)
2-17-2023	Ground	Aerodynamics	Test antenna drag force.
2-17-2023	Ground	Manufacturing	Test strength of antenna adapter.
2-21-2023	Ground	Performance	Ground mission test.
Final Aircraft (Planned)			
3-26-2023	Flight	Performance	M1 simulated flight. Evaluate performance.
3-26-2023	Flight	Performance	M2 simulated flight. Evaluate performance.
3-26-2023	Flight	Performance	M3 simulated flight. Test deployment. Evaluate performance.
4-2-2023	Flight	Performance	Practice for competition.
4-9-2023	Flight	Performance	Practice for competition.

8. Performance Results

8.1 Ground Test Results

8.1.1 Propulsion Test Results

Results from the static thrust test confirmed that the propulsion system generally met the desired performance requirements. Test data is compared to eCalc predicted performance in the table below.

Table 8.1: Predicted vs. Actual Performance

Parameter	Prediction	Test Result
RPM	10534	10843
Thrust [Kgf]	4.128	3.598
Max Current [A]	46.89	54.7
Electrical Power [W]	980.5	1238

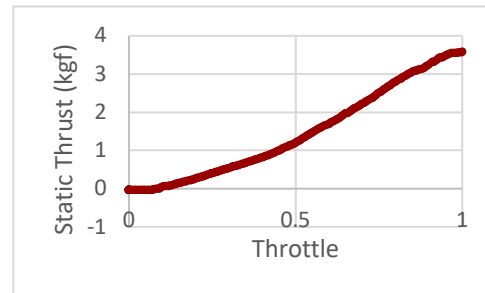


Figure 8.1: Static Thrust Test Results

Based on the static testing, the motor drew slightly more burst current than predicted. This is not an issue, however, since the ESC was sized with significant margin. The efficiency of the actual propulsion system appeared to be slightly lower than predicted, however the static thrust test was conducted indoors, and obstructions and ground effects may have contributed to that decrease in efficiency. Later flight testing confirmed that the propulsion system delivered the desired takeoff capability and cruise performance.

8.1.2 Bending Test Results

The Bending test showed that birch ply was the strongest material, followed by lite ply and then the two thicknesses of balsa. Fiberglassing increases the strength of all materials, most notably the plywoods. The fiberglassed 1/8 balsa had the highest strength-to-weight ratio. These data suggest that lite ply should be used in areas of moderate stress, such as the fuselage formers, and birch ply should be used in high stress locations, such as the motor mount. Balsa wood is still suitable for low stress locations but is not strong enough to replace many structural components. Charts of bending strength and specific bending force are shown below.

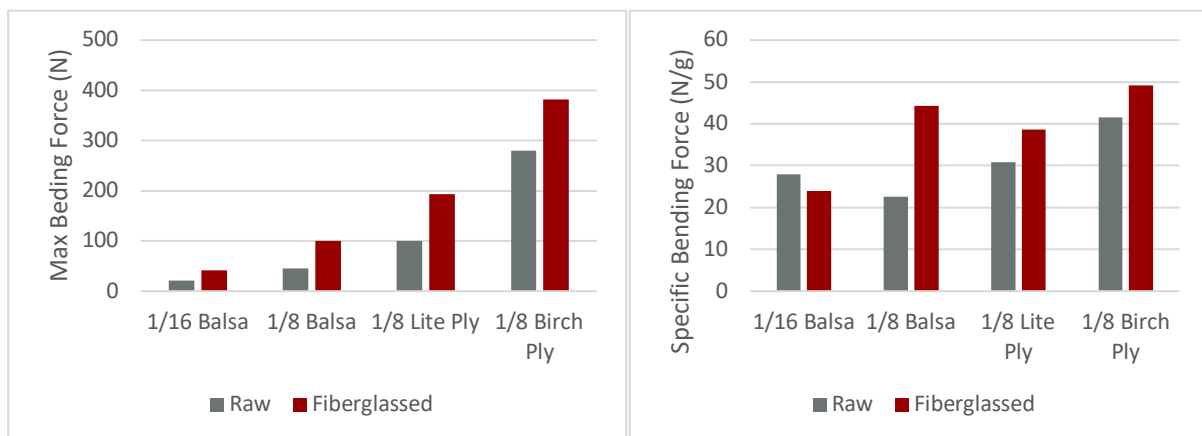


Figure 8.2: Plots of Bending Strength and Specific Bending Force

8.1.3 Wingtip Test Results

The aircraft was supported successfully by just the wingtips. The measured deflection was around 12 mm in each wingtip. This agrees with the FEA performed on the CF rod alone in Section 5.5.2 (predicted deflection of 15.7 mm) since the additional structural elements in the wing provide additional bending support.

8.1.4 Ground Mission Test Results

The mockup supported the predicted 60 kg load without breaking. The weight application method was also successful. Further testing of the final aircraft will verify how much weight can be added without deforming the other components such as the wings or the rest of the fuselage.

8.1.5 Antenna Test Results

The plot of drag force vs. velocity is shown in Figure 8.3. At 31 m/s, the drag force is approximately 10.5N. This is higher than the predicted value of 8 N, likely due to the vortex effects at the open end of the antenna. However, it is still within the acceptable stability limit.

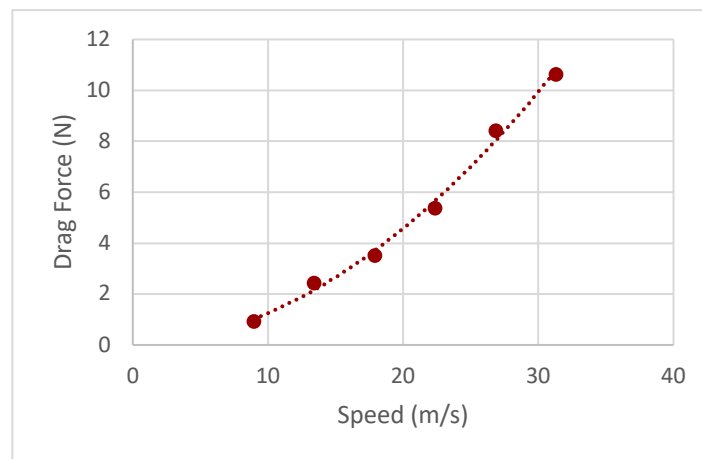


Figure 8.3: Plot of Drag Force vs. Speed

8.1.6 Adapter Test Results

A force was applied to the end of the antenna to simulate a 10.5N load at the adapter-antenna interface. The adapter and the wingtip successfully supported the load without breaking.

8.1.7 Box Test Results

The foam mockup consisted of the fuselage and empennage. The wings were not included since they are detachable, but the space on either side of the fuselage was sufficient to store two sets of wings. The mockup fit inside the shipping box without issue.

8.2 Flight Testing

8.2.1 Jan. 28th, 2021 – Shakedown Flight (Prototype)

The aircraft successfully took off within the 60ft takeoff limit. However, several minutes into the flight, the aircraft entered an uncontrollable spin and crashed. The cause was determined to be the high wing loading, which meant the pilot was required to fly at high speeds to avoid a stall. The ailerons were also far too sensitive to the pilot's inputs. The combination of these characteristics meant that when the aircraft became unstable, it was impossible to rectify its course without serious overcorrection. This failure motivated a number of design modifications, such as setting a wing loading limit during the preliminary design phase and adjusting the materials used during manufacturing. Table 8.2 compares the predicted performance with that of the prototype

Table 8.2: Predicted Performance versus Actual Performance

Flight Description	Predicted	Actual	Difference
M2 Cruise Velocity [m/s]	37.7	44.4	+17.77%
M3 Battery Endurance [min]	9.8	8.6	-12.2%
TOFL (ft)	32.57	45	+38.2%

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